REGIME-SWITCHING RISK MODELS FOR ETF PORTFOLIOS: DO MARKET-STATE SHIFTS IMPROVE TAIL-RISK FORECASTING?

by

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ABSTRACT

When the stock–bond correlation turned positive in 2022, portfolios of equities and Treasuries lost part of their hedge. I ask whether a risk manager forecasting one-day value-at-risk (VaR) and expected shortfall (ES) for an equal-weight portfolio of SPY, IEF and GLD should let the SPY–IEF correlation switch with a two-state market regime. The switching model and a constant-correlation model share the same GARCH margins and are scored with the FZ0 joint loss. The test, its decision rule, a smallest effect of interest and its simulated power were fixed before the 2020–2025 hold-out was opened, and the test was run once. Over 1,508 hold-out days, switching lowered the average loss by 0.0013 (95% interval -0.0037 to $+0.0012$; one-sided $p = 0.158$). The test could detect only differences above 0.0031, so a useful gain is neither shown nor ruled out, and the favourable part of the difference came in 2021 rather than in the 2022 reversal. A larger problem is shared by both models: each breached its 5% VaR on about 6.8% to 6.9% of days, and breach-day losses exceeded the forecast ES. A simple EWMA, scored post hoc, breached on 5.2% of days with a loss that could not be distinguished from either model's. On development data, regime information helped through volatility rather than correlation. For this portfolio the evidence gives no reason to adopt regime-switching correlation, and the scale of the forecast matters more than the correlation model.

KEYWORDS: Value-at-Risk; Expected Shortfall; Regime Switching; Stock–Bond Correlation; Forecast Evaluation; Statistical Power

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List of abbreviations and notation

Abbreviations

CCC	constant conditional correlation model
A, B, D	static covariance models: A is a pooled normal, B has regime-specific volatilities, D has regime-specific volatilities and correlations (Chapter 4)
DCC	dynamic conditional correlation model
DM	Diebold–Mariano test
ES	expected shortfall
ETF	exchange-traded fund
EWMA	exponentially weighted moving average (RiskMetrics variance forecast)
FZ0	the Fissler–Ziegel-type joint loss for VaR and ES used here (Chapter 5)
GARCH	generalised autoregressive conditional heteroskedasticity
GW	Giacomini–White test of conditional predictive ability
HAC	heteroskedasticity- and autocorrelation-consistent (variance estimator)
HMM	hidden Markov model
HS	historical simulation
MDE	minimum detectable difference
PSD	positive semi-definite
QML	quasi-maximum likelihood
RSC	regime-switching correlation model (all correlations switch)
RSC_P	regime-switching correlation model in which only the SPY–IEF correlation switches (the primary model)
SPY, IEF, GLD	the three funds of the portfolio: US equities, US Treasuries (7–10 years), gold
SESOI	smallest effect of interest
VaR	value-at-risk

Notation (full definitions at first use in Chapters 4 and 5)

t	forecast day; the forecast for day t uses returns through day $t-1$
y_t	portfolio simple return on day t
r_t	vector of the three asset returns (SPY, IEF, GLD)
w	portfolio weights, $(1/3, 1/3, 1/3)$, constant
α	tail level of the VaR and ES forecasts (5%, secondary 2.5%)
VaR_t, ES_t	forecast value-at-risk and expected shortfall of y_t (negative numbers)
S_t	hidden regime, 1 calm, 2 stress
d_t	loss difference, richer minus baseline: $d_t = L_t^{\mathrm{rich}} - L_t^{\mathrm{base}}$; negative favours the richer model
$\bar{d}$, $\widehat{\mathrm{SE}}$	mean loss difference and its HAC standard error
T	number of forecast days in an evaluation sample

Chapter 1: Introduction

A risk manager who reports a daily value-at-risk (VaR) and expected shortfall (ES) for a portfolio of stocks, Treasuries and gold has to say how those assets move together in a bad week. The correlation between stock and bond returns has no fixed sign. It was about 60% in the mid-1990s and about −60% in the early 2000s (Baele et al., 2010), and it switched from positive to negative around 2001 (Campbell et al., 2020). It turned positive again in 2022: the correlation of daily SPY and IEF returns was −0.46 over 2010 to 2019 and +0.19 in 2022, and it stayed near zero in 2023 (+0.04) and 2024 (+0.07) (Figure 3.1). A forecast that holds the correlation constant would be too optimistic in the tail on exactly the days when the bond holding stops offsetting equity losses. The stakes go beyond internal reporting: the Basel Committee's market-risk standard sets capital for internal models on expected shortfall at the 97.5% level, a 2.5% tail (Basel Committee on Banking Supervision, 2019), which is the secondary tail level used here.

The practical decision is whether a risk desk should replace a constant-correlation forecast with one in which the correlation switches with the market state. This thesis asks whether letting the SPY–IEF correlation switch with a market regime improves one-day-ahead VaR and ES forecasts of a fixed portfolio, an equal-weight holding of SPY, IEF and the gold fund GLD. The confirmatory comparison was fixed before I used the test sample: a model in which the SPY–IEF correlation depends on the state of a two-state hidden Markov model fitted to SPY returns, against a model with constant correlation and the same GARCH margins. A wider ladder of models, run on development data only, shows where in the chain of models regime information helps.

The economic reasoning comes from the stock–bond literature. Campbell et al. (2020) tie the sign of the correlation to a slowly changing macroeconomic state, while Baele et al. (2010) find that macroeconomic fundamentals explain little and that market conditions such as liquidity and the variance premium matter more. A two-state volatility regime is a market-based signal,

so the test speaks to the second reading. If that reading is right, the switching model should gain most when the recent stock–bond correlation is positive, a prediction that Chapter 7 checks. Because the model learns the stress-day correlation from the preceding five years, the test asks whether it improves forecasts, not whether it reproduces the positive correlation of 2022.

The literature leaves the question open (Chapter 2). Regime switching improves volatility-based risk forecasts (Ardia et al., 2018), regime-switching correlation has been proposed (Pelletier, 2006) and tested for densities and VaR mainly on stocks (Paolella et al., 2019), and univariate and multivariate models have been compared for expected shortfall without regime switching (Fortin et al., 2023). To my knowledge, no study of regime-switching correlation scores VaR and ES jointly for a stock–bond portfolio and reports what its test could and could not detect.

Switching the SPY–IEF correlation lowered the average loss by 0.0013 over the 1,508 hold-out days of 2020 to 2025, with a 95% interval from -0.0037 to $+0.0012$, and the pre-specified one-sided test did not reject ($p = 0.158$). A loss difference of that size is roughly the penalty for scaling both the VaR and ES forecasts 2% to 3% away from the truth (Section 5.3.3). The test could detect a difference of 0.0031 with 80% probability, and a true improvement equal to the smallest effect of interest, 0.0020, with probability about 48%. Six years of daily data therefore cannot separate a gain of that size from zero. The favourable part of the difference accrued in 2021, not in 2022, the year of the reversal (Figure 7.1).

For a risk desk the more urgent finding is that neither model was calibrated. Both breached the 5% VaR on about 6.8% to 6.9% of hold-out days, and losses on breach days exceeded the forecast expected shortfall. A RiskMetrics EWMA of the portfolio return, scored post hoc on the same days, breached on 5.2% with a loss indistinguishable from theirs (Section 7.4). On the development days, regime information helped through volatility rather than correlation.

Forecasts are scored with the FZ0 loss of Patton et al. (2019), which scores VaR and ES jointly, and models are compared with a Diebold–Mariano test with a heteroskedasticity- and

autocorrelation-consistent standard error. The sample is split into development days (2010–2019), on which I explored, and a hold-out (2020–2025), used once for one test whose decision rule, smallest effect of interest and simulated power were fixed in advance.

The thesis contributes a pre-specified estimate, with its detection limit, of whether regime-switching correlation improves joint VaR–ES forecasts for a stock–bond portfolio, and a decomposition of where regime information helps. Chapter 2 reviews the literature; Chapters 3 to 5 set out the data, models and evaluation; Chapters 6 to 9 report, discuss and conclude; and Chapters 10 and 11 document the code and reflect on the project.

Chapter 2: Literature Review

This chapter positions the thesis against the work closest to it.

2.1 Regime switching in risk forecasting

2.1.1 Regime-switching volatility and risk forecasts

Markov switching goes back to Hamilton (1989), and regime shifts matter for asset allocation (Ang and Bekaert, 2002), where regime-switching correlations and volatilities rise in bad times. For stocks and bonds together, Guidolin and Timmermann (2006) find that two or three states describe each asset's returns alone but that four states, which they label crash, slow growth, bull and recovery, are needed for the joint distribution. The regimes that matter for one asset need not be those that matter for its comovement with another. For risk forecasts, Ardia et al. (2018) report that Markov-switching GARCH models give more accurate VaR, ES and left-tail forecasts than single-regime counterparts, especially for stock returns, and that the improvement is larger when switching is added to simple specifications such as a normal GARCH. They score with the loss of Patton et al. (2019), which I also use. Their object is the volatility of a single return series. Whether letting the dependence between assets switch adds anything is a different question, and it is the one this thesis asks.

2.1.2 Gains relative to the strength of the baseline

The size of a regime gain depends on what it is measured against. Ardia et al. (2018) find the gain largest over simple baselines. In the multivariate literature, Fortin et al. (2023) find that static multivariate models tend to be dominated by univariate and dynamic multivariate ones, and that dynamic multivariate models are mostly indistinguishable from a univariate skewed-t GARCH-type model (Section 2.2.1 describes their design). A single-regime GARCH model is therefore a strong baseline, and the confirmatory comparison of Chapter 4 gives both models the same GARCH margins, so that a gain cannot be credited to a weak benchmark.

2.2 Stock–bond correlation: models and drivers

2.2.1 Correlation modelling and regime-switching correlation

Correlation can be held constant (Bollerslev, 1990), driven by a GARCH-type recursion (Engle, 2002), or switched between states. Pelletier (2006) proposes the last: a Markov chain selects one of several correlation matrices, each constant within its regime, combined with separate models for the standard deviations, and two-step estimation separates the correlation parameters from the volatility ones. Applied to four exchange rates and compared with a DCC model by in-sample likelihood and a non-nested test, the regime model fits better and gives smoother correlations; there is no out-of-sample VaR or ES evaluation. My model RSC is of the same family, with two differences: the regime signal is a hidden Markov model fitted to SPY returns, not a chain estimated jointly with the correlations, and the confirmatory version switches a single correlation and not the whole matrix.

The closest design is that of Paolella et al. (2019). In its working-paper version (Swiss Finance Institute Research Paper 19-27) the model has GARCH(1,1) margins that do not depend on the regime, regime-specific correlation matrices, and a forecast that mixes the regime densities with predicted regime probabilities. On 29 Dow Jones stocks it gives better density forecasts than constant correlation, while VaR accuracy is reported to depend largely on the heavy tails of the return distribution rather than on the correlation dynamics. The published version uses a multivariate generalised hyperbolic distribution and concludes that Markov-switching correlation gives accurate risk forecasts and could lower capital requirements in distressed periods. Neither version scores VaR and ES with a joint loss or studies a stock–bond portfolio.

The evaluation closest in method is Fortin et al. (2023), which compares three univariate with eight multivariate copula models, several with dynamic correlation, for weekly VaR and ES of portfolios of equity factors and of stocks. It scores the forecasts with the joint loss of Patton et al. (2019) and compares them with the conditional predictive ability test of Giacomini

and White (2006) and the model confidence set of Hansen et al. (2011). Two things are left open that I take up: the study does not use regime-switching correlation, and it reports losses, p-values and model sets without an interval for the loss difference or a power calculation, so a non-significant result is not shown to be small.

2.2.2 The sign and drivers of the stock–bond correlation

The stock–bond correlation has changed sign. Baele et al. (2010) document realised correlations of about 60% in the mid-1990s and about -60% in the early 2000s, and Campbell et al. (2020) report that the correlation of bond and stock returns was positive on average before the second quarter of 2001 and negative afterwards. The two studies disagree about the drivers. Campbell et al. (2020) link the switch to a change in the sign of the correlation between inflation and the output gap, dated to 2001 with a break test on macroeconomic data, and show in a habit-formation model that time-varying risk premia amplify the change. Baele et al. (2010) study a dynamic factor model with ten state variables and find that macroeconomic fundamentals explain little of the stock–bond correlation, that liquidity variables matter more, and that negative correlations arise when exposures depend on the variance premium. Neither study forecasts risk, and both end before 2012. Evidence that covers the recent reversal points back towards the macroeconomy: using long US and UK histories, Molenaar et al. (2024) find inflation, real rates and government creditworthiness to be important explanatory variables.

Two points follow. The correlation changes sign over multi-year spells, which motivates letting it switch. But whether the state behind the change is a slow macroeconomic one or a faster market-based one is not settled, and this bears on the choice of regime signal. Connolly et al. (2005) relate the stock–bond return correlation to stock-market uncertainty, measured by implied volatility, which fits the market-based reading. A two-state model of SPY volatility ties the correlation regime to equity-market stress in the same way.

2.3 Evaluating forecasts and reporting null results

2.3.1 Forecast evaluation and tests of predictive ability

Joint elicitability of VaR and ES (Fissler and Ziegel, 2016) underlies the FZ0 loss of Patton et al. (2019), and Nolde and Ziegel (2017) argue that such scoring functions should be used for comparative backtests alongside coverage tests (Kupiec, 1995; Christoffersen, 1998), the combination used here. Equal predictive ability is tested by Diebold and Mariano (1995) and, for forecasting methods with estimated parameters, by West (1996) and Giacomini and White (2006), whose setting this thesis adopts. Nested comparisons raise further issues (Clark and McCracken, 2001; Clark and West, 2007; McCracken, 2020), addressed in Chapter 5; superior predictive ability across several candidates is tested by Hansen (2005).

2.3.2 Negative results, power and pre-registration

Hoenig and Heisey (2001) argue that power calculated after the experiment to interpret a non-significant result is fundamentally flawed, which is why Chapter 5 reports power only for stated effect sizes. Abadie (2020) argues that non-significant results can be highly informative and should be reported visibly, and Ioannidis et al. (2017) report that the median power across 159 economics literatures is 18% or less. Lakens (2017) proposes equivalence tests with bounds set by the smallest effect of interest, the idea behind outcome O3, and Nosek et al. (2018) describe pre-registration as separating prediction from postdiction.

2.4 What this thesis adds

The thesis combines three elements that the studies above treat separately: a ladder of models that separates the treatment of volatility, correlation and the mean under one regime signal; VaR and ES forecasts of a fixed stock–bond–gold portfolio scored with a joint loss, with the test, its sidedness and its decision rule fixed before the test sample was used; and minimum detectable differences and simulated power reported next to every result. The confirmatory question is whether letting the stock–bond correlation switch with an equity-volatility regime

improves tail-risk forecasts over constant correlation, with the same GARCH margins in both models, in a period that contains the 2022 reversal.

Chapter 3: Data and Sample Design

This chapter describes the data and portfolio, the patterns that motivate the analysis, and the split into development and hold-out periods.

3.1 Data

3.1.1 The main sample

The main sample is three exchange-traded funds: SPY (US large-cap equities), IEF (US Treasuries with seven to ten years to maturity) and GLD (gold). The data are daily adjusted closing prices from Yahoo Finance, obtained through the `yfinance` package; the data vintage is 26 September 2026, and the file and its SHA-256 hash are stored with the code (Chapter 10). The estimation code truncates the series at 31 December 2025 before computing anything.

I compute simple daily returns, $r_{i,t} = P_{i,t}/P_{i,t-1} - 1$, from the adjusted close. Dates on which any of the three prices is missing are dropped and returns are computed between consecutive available dates, with no forward filling. Through 31 December 2019 the file yields 3,805 prices and 3,804 returns, the first on 19 November 2004. Adjusted closes include distributions, so a return measures the total return of holding the fund. Daily ETF prices are the data a desk forecasting this portfolio would itself use, and they cover the 2022 reversal; their weakness, examined below, is that they start too late to contain an earlier spell of positive stock–bond correlation.

3.1.2 Additional samples used in the development period

The ladder of comparisons in Chapter 6 is also run on three further ETF samples: SPY, IEF and HYG; four sector funds (XLU, XLK, XLF, XLE); and four risk assets (HYG, EEM, VNQ, DBC). They were used for a check declared before it was run (Section 4.4). The first contains a Treasury fund, whose correlation with equities falls in stress in the development data; in the other two the correlations rise in stress. HYG has a shorter history, so the two samples that

Table 3.1: Summary statistics of daily returns

	Mean (% p.a.)	Volatility (% p.a.)	Skewness	Excess kurtosis	Minimum (%)	5% quantile (%)
Development, 2010–2019 (2,516 days; SPY–IEF correlation −0.46)						
SPY	13.7	14.7	−0.42	4.4	−6.51	−1.52
IEF	4.5	6.0	−0.13	1.1	−1.68	−0.58
GLD	4.1	15.5	−0.47	5.3	−8.78	−1.49
Portfolio	7.4	7.1	−0.41	4.2	−3.63	−0.70
Hold-out, 2020–2025 (1,508 days; SPY–IEF correlation −0.08)						
SPY	16.2	20.7	−0.27	13.0	−10.94	−1.81
IEF	0.4	7.6	0.22	2.6	−2.51	−0.75
GLD	18.4	16.3	−0.36	3.0	−6.43	−1.63
Portfolio	11.6	10.0	−0.02	7.3	−4.50	−0.95

Note: Simple daily returns from adjusted closing prices; mean ×252, volatility $\times\sqrt{252}$. Portfolio: equal weights, rebalanced daily. The 5% quantile is the empirical counterpart of the 5% VaR. Source: `output/tables/data_summary.json`, from `scripts/thesis_data_summary.py`.

contain it start forecasting in 2012. A fifth sample uses two mutual funds as proxies for stocks and long Treasuries (VFINX and VUSTX, equally weighted), with forecasts from 1992 to 2019, to include a period in which the stock–bond correlation was positive. The funds are not ETFs, and VUSTX tracks long Treasuries and not IEF. None of these samples is used in the hold-out, and their data stop at 31 December 2019.

3.2 The portfolio

The portfolio holds the three funds with constant weights $w = (1/3, 1/3, 1/3)$. The portfolio return is $y_t = w' r_t$. Holding the weights constant means that the portfolio is rebalanced to equal weights every day. I do not model the turnover this implies, nor transaction costs, so the portfolio is a clean object on which to forecast risk and not an investable strategy. A buy-and-hold portfolio would have drifting weights and would give different numbers; every result in this thesis uses constant weights.

3.3 Summary statistics and the motivating pattern

Table 3.1 summarises daily returns in the two forecast periods, and three features of it shape the analysis. First, the stock–bond hedge weakened. The SPY–IEF correlation was −0.46 over 2010–2019 and −0.08 over 2020–2025, and the 63-day rolling correlation, plotted in Figure 3.1,

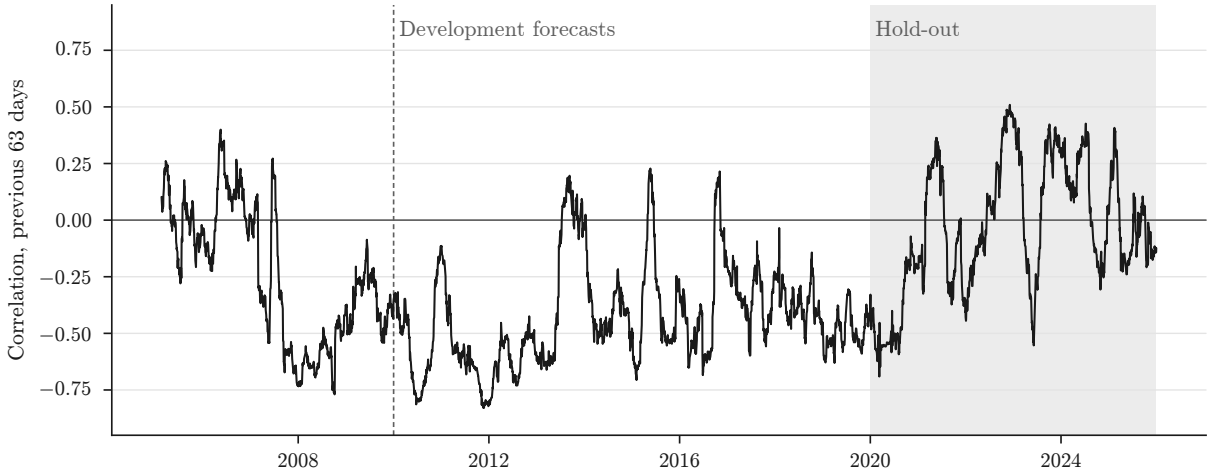


Figure 3.1: Correlation of daily SPY and IEF returns over the previous 63 trading days, 2005–2025

Note: Correlation of simple daily returns over a rolling 63-day window. The dashed line marks the first development forecast (4 January 2010); the shaded area is the hold-out (2020–2025). Source: `scripts/thesis_data_summary.py`.

Table 3.2: Share of days with a positive 252-day stock–bond correlation, by era

Era	Share of days
1987–1999	90.6%
2000–2009	23.7%
2010–2019	0.0%

Note: Mutual-fund proxies for stocks and long Treasuries (VFINX, VUSTX), not ETFs. Source: `output/explore/longhistory_summary.json`.

was positive on 6.4% of development days against 44.2% of hold-out days. Second, the portfolio became riskier and more fat-tailed: its annualised volatility rose from 7.1% to 10.0% and its excess kurtosis from 4.2 to 7.3. A model estimated on the development years therefore starts the hold-out with a calm view of risk. Third, every series has fat tails in both periods, which is why a Student-t variant is reported next to the Gaussian forecasts.

A longer history puts the development period in context. Table 3.2 uses two mutual funds as proxies for stocks and long Treasuries: the 252-day stock–bond correlation was positive on 90.6% of days in 1987–1999, on 23.7% in 2000–2009 and on 0.0% in 2010–2019. No ETF development sample contains the regime the thesis is about, so the development data cannot show whether a correlation-switching model detects it. Only the hold-out can.

Table 3.3: Sample periods and their status

Period	Forecast days	Days	Status
Development	4 Jan 2010 – 31 Dec 2019	2,516	Specifications were chosen after seeing results on these days.
of which early	4 Jan 2010 – 31 Dec 2015	1,510	Same status as the development period.
of which late (dry run)	4 Jan 2016 – 31 Dec 2019	1,006	Used to test the pipeline. Not an independent validation.
Hold-out	2 Jan 2020 – 31 Dec 2025	1,508	One run of one pre-specified test, after the specification is fixed.

Note: ETF samples that contain HYG start forecasting in 2012 and have 1,944 development days. The first estimation window for the first hold-out forecast covers 30 December 2014 to 31 December 2019. Day counts from the result files `ladder_dm_table.csv` and `routes_dm_table.csv`; the hold-out count is a count of dates in the raw file.

3.4 Sample design

3.4.1 Development and hold-out periods

Table 3.3 sets out the periods. Each forecast for day t uses the 1,260 returns before t (Section 4.1), so every sample also needs five years of earlier returns.

The development period is where I explored. I screened ten ETF universes, ten regime definitions and two tail levels, plus the long-history sample. The headline family contains 214 tests of one comparison (D versus B, defined in Chapter 4); every specification tried is logged and none was dropped. Results on the development period are therefore descriptive. Anything chosen because it looked good there, including the confirmatory test itself, has no claim to be validated by those same days.

The late part of the development period (2016–2019) serves as a dry run: I ran the final code path on it to check the pipeline end to end, and its estimates enter no argument about the hypothesis. The hold-out is 2 January 2020 to 31 December 2025, 1,508 forecast days. It contains the 2020 crash, the 2022 stock–bond reversal that motivates the thesis and the years around it. I ran the pre-specified test on it once, after the specification was fixed (Section 5.5).

3.4.2 What was known before the specification was fixed

No model was fitted or tuned on 2020–2025, but two things that bear on the result were known. I knew that the stock–bond correlation turned positive in 2022, and that knowledge motivated the question. And the confirmatory test was chosen from simulated power on a process in which the SPY–IEF correlation shift is largest by construction (Section 5.3.2). The specification, scoring code and raw-data hash were fixed in the git tag `spec-freeze-v1` before the run. Appendix A lists the remaining, minor items seen before the freeze and the one amendment made to the scorer before any scoring.

Chapter 4: Methodology

The tail risk of a portfolio depends on how volatile each asset is and on how the assets move together, and a market regime can change either. The models in this chapter share one regime signal, one portfolio and one estimation scheme. They differ in whether volatility, correlation or both depend on the regime, and in whether volatility also follows GARCH dynamics. Pairing them so that each comparison changes as little as possible (Section 4.4) is what lets the thesis ask about the correlation channel specifically. Chapter 5 defines the loss, the test and the decision rule; Appendix C gives the longer formulas and Chapter 10 the implementation settings.

4.1 The forecasting task and the estimation scheme

For each forecast day t I forecast the one-day-ahead value-at-risk VaR_t and expected shortfall ES_t of the portfolio return y_t at tail level α, using returns through day $t-1$ only. VaR_t is the α-quantile of the forecast distribution of y_t, and ES_t is the mean of y_t below that quantile. Both are negative numbers. The primary tail level is $\alpha = 5\%$ and the secondary one is 2.5%.

Forecasts come from a walk-forward scheme. The forecast days are split into blocks of 21 consecutive days. For the block starting at day t_0, every model is estimated on the rolling window of the 1,260 returns before t_0 (about five years) and used, without re-estimation, for the 21 days of the block. The regime model is refitted at every block. The GARCH margins of Section 4.3.2 are refitted every third block, that is every 63 days; at the other blocks their parameters are kept and only the window is re-filtered. A model is therefore a forecasting procedure, consisting of the specification, the window length and the refit schedule, and this is the object that Chapter 5 compares.

4.2 The regime signal

The regime signal is a two-state Gaussian hidden Markov model (HMM) fitted to SPY daily returns only, so that every model uses the same signal. The hidden state $S_t \in \{1, 2\}$ is calm (1, lower return variance) or stressed (2). Given $S_t = k$, the SPY return is normal with mean

μ_k and variance σ_k^2, and S_t follows a Markov chain with transition matrix P. The parameters are estimated on each window by the expectation–maximisation algorithm, and the states are ordered by variance after each fit, so the stress state keeps its label.

Two sets of probabilities are used, and they must not be confused. Inside the window, the smoothed probabilities $\gamma_{s,k} = \Pr(S_s = k \mid \text{window})$ for each window day s weight the regime moments below; they use no data beyond the end of the window. For the forecast, the weights of the mixture are the predictive probabilities

$$p_{t,k} = \Pr\big(S_t = k \mid \text{SPY returns through } t-1\big), \tag{1}$$

obtained by propagating the filtered probability at the window end through P and, within the block, filtering with the realised SPY returns up to day $t-1$. Day t's own return is never used.

4.3 The forecasting models

4.3.1 Static regime-covariance models: A, B and D

The static models use weighted sample moments of the asset returns over the window (Appendix C): pooled moments $\hat{\mu}, \hat{\Sigma}$ with equal weights, and regime moments $\hat{\mu}_k, \hat{\Sigma}_k$ weighted by $\gamma_{s,k}$. Write $\hat{\Sigma} = \operatorname{diag}(\hat{\sigma})\hat{R}\operatorname{diag}(\hat{\sigma})$ and $\hat{\Sigma}_k = \operatorname{diag}(\hat{\sigma}_k)\hat{R}_k\operatorname{diag}(\hat{\sigma}_k)$ for their decompositions into standard deviations and correlations. The three static models assign to regime component k the covariance

$$\text{A (pooled normal):} \quad \Sigma_k^A = \hat{\Sigma}, \tag{2}$$

$$\text{B (regime volatility):} \quad \Sigma_k^B = \operatorname{diag}(\hat{\sigma}_k)\,\hat{R}\,\operatorname{diag}(\hat{\sigma}_k), \tag{3}$$

$$\text{D (regime volatility and correlation):} \quad \Sigma_k^D = \hat{\Sigma}_k. \tag{4}$$

Model A has the same covariance in both components, so it is a single normal distribution. Model B lets the volatilities differ by regime and holds the correlation matrix at its pooled value.

Model D lets every volatility and every correlation differ by regime. The portfolio variance in component k is $s_k^2 = w'\Sigma_k w$, and every component has the same mean $m = w'\hat{\mu}$, the pooled window mean (the "pooled" mean treatment used in the ladder tables). The covariances are fixed within a 21-day block; only the mixture weights $p_{t,k}$ move from day to day. Two controls vary the mean: a regime-specific mean ($w'\hat{\mu}_k$ in component k) and a zero mean. Appendix B reports them, because they change the size of some contrasts.

4.3.2 Models with GARCH margins: CCC, RSC and RSC_P

These models replace the block-constant volatilities by GARCH(1,1) volatilities that update daily. For each asset i, with returns in per cent, $x_{i,s} = 100\, r_{i,s}$, I estimate on the window

$$x_{i,s} = \mu_i + \varepsilon_{i,s}, \qquad \sigma_{i,s+1}^2 = \omega_i + a_i\, \varepsilon_{i,s}^2 + b_i\, \sigma_{i,s}^2, \tag{5}$$

by Gaussian quasi-maximum likelihood, one asset at a time. The fitted recursion gives the standardised residuals $z_{i,s} = \varepsilon_{i,s}/\sigma_{i,s}$ in the window. Within the block, each volatility is updated daily from the realised return with the parameters held fixed (Chapter 10). Write $A_t = w \circ \sigma_t / 100$ for the vector of weighted volatilities, where $\circ$ is the element-wise product. The volatility is the same in every regime component, so the regime enters only through the correlation matrix of z and the mixture weights.

The three models differ in the correlation matrix of the standardised residuals:

$$\text{CCC:} \quad R_1 = R_2 = \bar{R}, \text{ a single component;} \tag{6}$$

$$\text{RSC:} \quad R_k = \bar{R}_k; \tag{7}$$

$$\text{RSC_P:} \quad R_k = \bar{R} \text{ with the (SPY, IEF) entry replaced by } (\bar{R}_k)_{12}. \tag{8}$$

Here $\bar{R}$ is the correlation matrix of z over the window with equal weights and $\bar{R}_k$ is the correlation matrix of z with weights $\gamma_{s,k}$. In RSC_P only the SPY–IEF correlation switches with the regime

and the other two correlations stay pooled. I fixed this pair in advance from the stock–bond hypothesis, not by scanning. A repair that keeps every matrix a valid correlation matrix is in place but never fired (Chapter 10). The forecast distribution of y_t is a mixture of normals with weights $p_{t,k}$ ($p_{t,k} = 1$ for CCC) and component variances $A_t' R_k A_t$. Its mean, $m = w'\hat{\mu}^{\mathrm{QML}}/100$ from the QML constants, is common to all components and to both models of the primary test.

4.3.3 Mixture value-at-risk and expected shortfall

For a forecast mixture with weights p_k, means m_k and standard deviations s_k, the value-at-risk is the solution q of

$$\sum_k p_k \, \Phi\!\left(\frac{q - m_k}{s_k}\right) = \alpha, \tag{9}$$

found by root-finding, where Φ is the standard normal distribution function. The expected shortfall is the probability-weighted mean of the return below q, which has a closed form (Appendix C). Both are returns, so both are negative at the levels used here. This is the Gaussian chain, which is the primary one. In the secondary Student-t chain every component is a Student-t distribution with common degrees of freedom ν, scaled to the component's standard deviation. Appendix C describes how ν is estimated.

4.3.4 Benchmarks

Two other models serve as reference points. The first is a dynamic conditional correlation model (DCC; Engle, 2002) built on the same GARCH(1,1) margins, in which the correlation matrix follows a GARCH-type recursion in the standardised residuals (Appendix C). Its parameters are re-estimated every 63 days and the recursion is updated daily; if the estimates are not admissible, it falls back to constant correlation (Chapter 10). The second is a GARCH(1,1) model with Student-t errors fitted by maximum likelihood to the portfolio return itself (GARCH-t), which uses no information about the individual assets. Two practitioner baselines, both on the portfolio return, were added only after the hold-out was scored (Section 7.4): historical simulation, the empirical 5% quantile and the mean of the returns below it over the last 500

days; and a RiskMetrics exponentially weighted variance with decay 0.94, a zero mean and a normal distribution.

4.4 What each comparison changes

The thesis compares models in pairs that share the data, the regime signal and the estimation window. I do not call these pairs a nested sequence, because several pairs differ in more than one respect. Table 4.1 lists, for each comparison, the richer model and the baseline, what differs between them, and whether one model is a special case of the other.

Three consequences follow. Only D versus B, RSC versus CCC and RSC_P versus CCC change the correlation matrix alone, so only they speak to the correlation channel, and only within the volatility model used. CCC versus B and RSC versus D change the volatility model, the correlation input, the mean estimator and the number of components together; they compare model classes and cannot be read as the effect of volatility dynamics, since the GARCH models also drop the regime dependence of volatility. And because CCC, RSC and RSC_P share their margins, the primary test, RSC_P versus CCC, differs in two respects only: a mixture of two components against one, and a regime-specific SPY–IEF correlation against a pooled one.

Two families of comparisons were declared before they were scored. The first is the volatility-model contrast RSC versus D with a pooled mean, in the four ETF samples, with Holm's adjustment over the four (a gate: the ladder would become the main thesis if the contrast were negative in all four samples and significant after adjustment in at least two). The second is the confirmatory test RSC_P versus CCC (Chapter 5).

Table 4.1: What differs between the two models in each comparison

Comparison (richer vs baseline)	Volatility	Correlation matrix	Mean	Components	Relation
B vs A	regime-specific block SDs vs one pooled SD	pooled raw-return correlation in both	pooled window mean	2 vs 1	nested in volatility
D vs B	same regime SDs	regime-specific vs pooled, raw returns	pooled window mean	same	nested in the correlation matrix
CCC vs B	single-regime GARCH(1,1), daily update, vs regime-specific block SDs	standardised residuals vs raw returns	QML constant vs window mean	1 vs 2	non-nested: changes four things
RSC vs D	single-regime GARCH(1,1) vs regime-specific block SDs	regime-specific; standardised residuals vs raw returns	QML constant vs window mean	2 and 2	non-nested: model-class comparison
RSC vs CCC	identical GARCH	regime-specific vs pooled, same z	identical	2 vs 1	nested in the correlation matrix
RSC_P vs CCC (primary)	identical GARCH	only the (SPY, IEF) entry regime-specific	identical	2 vs 1	nested in one correlation entry
DCC vs CCC	same specification, fitted separately; own variance initialisation	recursion updated daily vs one pooled matrix per block	identical	1 vs 1	parameter-nested ($\theta_1 = \theta_2 = 0$); different code path
DCC vs RSC	as above	smooth recursion vs regime-weighted	identical	1 vs 2	non-nested

Note: "Block SDs" means standard deviations estimated on the window and held fixed for 21 days. B reduces to A when the two regime SDs are equal; D reduces to B and RSC to CCC when the regime correlation matrices equal the pooled one. The DCC filter starts from the `arch` package's own variance initialisation and carries its correlation state forward daily, so it is not the CCC fit with a constant added. The primary comparison is in bold.

Chapter 5: Evaluation and Inference

5.1 The loss function

A forecast of $(\mathrm{VaR}_t, \mathrm{ES}_t)$ is scored against the realised return y_t with the loss

$$\mathrm{FZ0}(y_t, \mathrm{VaR}_t, \mathrm{ES}_t; \alpha) = -\frac{\mathbf{1}\{y_t \le \mathrm{VaR}_t\}\,(\mathrm{VaR}_t - y_t)}{\alpha\,\mathrm{ES}_t} + \frac{\mathrm{VaR}_t}{\mathrm{ES}_t} + \log(-\mathrm{ES}_t) - 1, \tag{10}$$

where $\mathrm{VaR}_t < 0$ and $\mathrm{ES}_t < 0$ are return quantiles. Lower is better. This is the member of the Fissler–Ziegel family (Fissler and Ziegel, 2016) that Patton et al. (2019) call FZ0. Joint elicitability means that the expected loss is smallest at the true VaR and ES pair, so a forecaster cannot lower the expected loss by misreporting either; this is the reason to use such a loss. Scaling the return and both forecasts by $c > 0$ adds $\log c$ to the loss, so a difference between the losses of two models does not depend on the units of the return. Section 5.3.3 translates a loss difference into a proportional error in the forecasts.

5.2 The confirmatory test

5.2.1 Estimand

Let model R (richer) and model B (baseline) be two forecasting procedures, each a model together with its window and refit schedule (Section 4.1). Define the loss difference on forecast day t as

$$d_t = \mathrm{FZ0}_t^R - \mathrm{FZ0}_t^B, \tag{11}$$

so that a negative d_t favours the richer model. This sign convention, richer minus baseline with negative meaning better, is used in every table. The estimand is the long-run mean

$$\mu = \mathbb{E}[d_t], \tag{12}$$

the expected loss difference between the two procedures, with the estimation error in their parameters included. It is not a statement about whether the extra parameters of the richer model are zero in a population. This is the sense of equal predictive ability in Giacomini and White (2006), who compare forecasting methods, including their finite estimation windows, so that the test applies whether or not one model nests the other (see also Diebold, 2015). Their unconditional test is the two-sided Diebold–Mariano statistic; the one-sided version used here is my adaptation. With a rolling window of fixed length, as here (1,260 returns), their asymptotic result remains valid; the counterexample of McCracken (2020) concerns a fixed estimation sample.

5.2.2 Estimator and test

The estimator of μ is the sample mean $\bar{d} = T^{-1} \sum_{t=1}^{T} d_t$ over the T forecast days. Its standard error $\widehat{\mathrm{SE}}$ comes from a Bartlett-kernel heteroskedasticity- and autocorrelation-consistent (HAC) estimator (Appendix C) with the lag $m = \lfloor 4\,(T/100)^{2/9} \rfloor$, the rule of thumb associated with Newey and West (1994); at $T = 1{,}508$ this gives $m = 7$. The test statistic is the Diebold–Mariano statistic (Diebold and Mariano, 1995),

$$t_{\mathrm{DM}} = \bar{d} / \widehat{\mathrm{SE}}, \tag{13}$$

compared with the standard normal distribution; the small-sample correction of Harvey et al. (1997) would change it by a factor of 0.9997 and is not applied. The null hypothesis and its alternative are statements about μ:

$$H_0 : \ \mu \geq 0 \qquad \text{against} \qquad H_1 : \ \mu < 0. \tag{14}$$

In words, under H_0 the richer model does not have a lower expected loss than the baseline. The one-sided p-value is $\Phi(t_{\mathrm{DM}})$ and the two-sided p-value is $2\{1 - \Phi(|t_{\mathrm{DM}}|)\}$. Every table gives the

two-sided 95% interval $\bar{d} \pm 1.96\,\widehat{\mathrm{SE}}$ and labels each p-value one- or two-sided.

Normal critical values need not be accurate when one model nests the other (Clark and McCracken, 2001; Clark and West, 2007), but those results concern linear models under squared-error loss and do not carry over to the mixture forecasts and loss used here. For a rolling window, Clark and West (2007) themselves read the usual t-statistic as a statement about the estimated-parameter mean difference, which is the reading used here. I therefore read the size of the test from simulation (Section 5.3.2) and not from asymptotic theory.

5.2.3 Decision rule and interpretation

The confirmatory test is RSC_P against CCC, at tail level $\alpha = 5\%$ with a Gaussian chain (Section 4.4). Two levels should not be confused: $\alpha = 5\%$ is the tail level of the forecast, and the test has its own size, also 5%. The 5% tail level was chosen for power, since a 5% VaR is breached about twice as often as a 2.5% one and gives the loss difference more information; the 2.5% level of the Basel expected-shortfall standard is the secondary level. The decision rule is one-sided: reject H_0 if $t_{\mathrm{DM}} < -1.6449$. I chose a one-sided test because the substantive question has one direction, and because the simulated power and size of Chapter 6 were computed for exactly this rule. A one-sided test says nothing about harm, so I declare a separate statement: the switching model is called significantly worse if $t_{\mathrm{DM}} > +1.96$. The threshold is 1.96 rather than 1.6449 because in the simulated null the upper-tail rate is 5.2% at 1.96 but 8.0% at 1.6449 (Table 6.6).

The four possible outcomes are checked in this order and each licenses one sentence, fixed before the run. The full wording of O1, O3 and O4 is in Appendix A.3.

O1 $t_{\mathrm{DM}} \leq -1.6449$: statistical evidence at the one-sided 5% level that switching lowers expected FZ0 loss.

O4 $t_{\mathrm{DM}} \geq +1.96$: evidence that the switching specification has higher expected FZ0 loss.

O3 otherwise, and $\bar{d} - 1.6449\,\widehat{\mathrm{SE}} > -0.002$: the data are inconsistent, at the one-sided 5% level,

with an improvement larger than SESOI = 0.002 FZ0 units.

O2 otherwise: "The hold-out data neither establish an improvement nor exclude one. The estimate is $\bar{d}$ (95% CI $[a, b]$); an improvement of at least M (the minimum detectable difference for the one-sided 5% test, 80% power) would have been detected with 80% probability, and smaller improvements, including none, are consistent with the data."

No outcome is worded as an absence of an effect. The interval and the minimum detectable difference are reported whatever the outcome.

5.3 What the test can detect

5.3.1 Minimum detectable difference and power

The minimum detectable difference (MDE) is the smallest true difference that a test detects with 80% probability, given its standard error. For the two-sided 5% test it is $(1.96 + 0.8416)\,\widehat{\mathrm{SE}} = 2.8016\,\widehat{\mathrm{SE}}$, and for the one-sided 5% test it is $(1.6449{+}0.8416)\,\widehat{\mathrm{SE}} = 2.4865\,\widehat{\mathrm{SE}}$. Every comparison table gives the two-sided value, labelled as such, and the primary test gives both. The MDE depends on the design (the sample length and the variability of the loss difference) and not on the estimate, so it can be reported next to a result. I never report power computed at the observed estimate: that quantity is a function of the p-value and adds no information (Hoenig and Heisey, 2001). Power is reported only for stated effect sizes, which are the simulated scenarios below, the smallest effect of interest and the MDE.

5.3.2 The power study

I simulate a stylised process with the structure of the problem and score the same pipeline on it. A two-state Markov chain drives an equity-like, a bond-like and a gold-like asset whose volatilities are higher in the stress state and also follow GARCH(1,1) dynamics, with Student-t innovations; Table C.1 in Appendix C gives the parameters. The four scenarios differ only in the stress-state correlations. S0 leaves them at their calm values and is the null for the correlation steps; S1 is a moderate breakdown, in which the equity–bond correlation rises from -0.30 to

0.00; S2 is a strong breakdown, to +0.40; and S3 is a flight to quality, to −0.55. Each replication simulates 1,260 burn-in days and 1,508 test days (the length of the hold-out), runs the whole pipeline of Chapter 4 including the HMM on the first asset, and applies the test of Section 5.2.2. There are 500 replications per scenario, with seeds distinct from an earlier study that used 2,520 test days. Size is the rejection rate in S0, and power is the rejection rate in S1 to S3, each with a binomial 95% interval.

This is an assumption and not a measurement. The parameters are chosen, not estimated from data, and the shift in stress correlation is largest in the stock–bond pair by construction. The choice of RSC_P as the confirmatory test rested on an earlier run of this simulation, in which switching only the SPY–IEF correlation had equal size and higher power than switching all correlations. The decision was made before the hold-out was opened and is not reopened here, whatever the 1,508-day study shows.

5.3.3 The smallest effect of interest

A non-significant result does not show that an effect is absent. To give a negative result some content I declare one reference value, the smallest effect of interest (SESOI), and use it as a one-sided bound. To express it in economic terms, suppose a forecaster's VaR and ES are both a factor c times the true values of a known distribution. The expected loss then exceeds that of the true forecast by a penalty roughly proportional to $(c-1)^2$; Table 5.1 gives it, computed by numerical integration with no market data (`scripts/confirm_04_sesoi.py`). For a normal at $\alpha = 5\%$, understating or overstating both forecasts by 3% costs about 0.0016 to 0.0018, so I declare SESOI = 0.002, the average of these two penalties to one significant figure. That a 3% error in a capital-relevant quantity is economically meaningful is my assumption and not a fact. Under a Student-t distribution with six degrees of freedom the penalty is about 20% lower and rounds to 0.001; I report this as a caveat and keep the declared value.

The reference value is used in outcome O3 only. With a standard error of 0.0010 to 0.0015, the range suggested by the development data, the one-sided bound $\bar{d} - 1.6449\,\widehat{\mathrm{SE}}$ lies 0.0021 to

Table 5.1: Expected FZ0 penalty of scaling both VaR and ES by a constant factor

True distribution	$c = 0.95$	$c = 0.97$	$c = 1.03$	$c = 1.05$
Normal, $\alpha = 5\%$	0.0051	0.0018	0.0016	0.0042
Student-$t(6)$, $\alpha = 5\%$	0.0040	0.0014	0.0013	0.0034
Normal, $\alpha = 2.5\%$	0.0067	0.0023	0.0020	0.0055

Note: Analytic expected loss at scale factor c minus the expected loss at $c = 1$, for a unit-variance true distribution; no market data are used. Source: `output/confirm/sesoi.json`.

0.0025 below the estimate, so O3 requires an estimate near zero or positive and is unlikely; I stated this in advance. A two-one-sided-tests equivalence procedure with margin 0.002 is out of reach, since the half-width of its 90% interval alone would exceed the margin, so the thesis does not claim equivalence under any outcome.

5.4 Calibration tests

Next to the loss I report, as description only, whether the forecasts have the right coverage. The hit rate, the share of days with $y_t \leq \mathrm{VaR}_t$, should be near α. The Kupiec (1995) test compares the two, the Christoffersen (1998) test asks whether hits cluster in time, and their sum is the conditional-coverage test. For expected shortfall I use

$$Z_2 = \frac{1}{T\alpha} \sum_{t=1}^{T} \frac{\mathbf{1}\{y_t \leq \mathrm{VaR}_t\}\, y_t}{\mathrm{ES}_t} - 1, \tag{15}$$

which is zero in expectation if the forecast distribution is correct and positive if risk is underestimated. It is the Z_2 of Acerbi and Székely (2014) with the sign reversed, because they write the expected shortfall as a positive loss, and its expectation does not require independent observations. Its p-value is the share of 5,000 samples, each drawn from the days' forecast mixtures, whose Z_2 is at or above the observed value.

5.5 Multiple testing and what is pre-specified

The confirmatory family has one member, so no adjustment is needed. Besides it I score six secondary analyses of the same forecasts, none of which adds a model, a sample or a regime signal: the six ladder steps at the hold-out; the calendar year 2022 (inter-

val only); a Giacomini–White conditional test of the primary pair with the instruments {1, lagged 63-day SPY–IEF correlation} (a Wald statistic with a χ^2_2 reference, read with caution because Zhu and Timmermann (2020), in a preprint, show that the conditional null can fail when forecasts rest on estimated parameters); the same comparison at $\alpha = 2.5\%$ and in the Student-t chain; a test of superior predictive ability against GARCH-t (Hansen, 2005; stationary bootstrap, 2,000 draws, seed 0; candidates B, D, RSC, RSC_P); and the calibration tests above. They are unadjusted and labelled secondary, and none is cited as confirmation. Anything else computed is marked POST HOC and carries no inferential weight. The multiple-testing adjustments of the development period (Holm, 1979) are described with those results (Chapter 6).

The specification (`docs/preregistration_confirmatory.md`) follows the logic of pre-registration (Nosek et al., 2018) within the limits stated in Section 3.4.2; Appendix A records the run and its one amendment.

Chapter 6: Results on the Development Data

This chapter reports what the development data (forecast days through 2019) show about where regime information helps, and how large an effect the design could reveal. In short, modelling volatility matters and switching correlations barely moves the loss. The results are descriptive (Section 3.4), and Table 4.1 states what each comparison changes.

6.1 How to read the tables

Tables report the richer-minus-baseline difference $\bar{d}$ (negative favours the richer model), its HAC standard error, the 95% interval, the two-sided p-value and the MDE, as defined in Chapter 5; a result inside the MDE is inconclusive, not evidence of no effect. For scale, the smallest effect of interest (Section 5.3.3) is 0.002.

6.2 The ladder of comparisons

6.2.1 Regime-specific volatility against a pooled normal

Giving each regime its own volatilities (model B) lowered the loss against the pooled normal A by 0.16 to 0.18 in three of the four ETF samples, about eighty to ninety times the smallest effect of interest, each with a two-sided p-value below 0.001, and by 0.11 in the fund sample (Table 6.1). The exception is the main sample, SPY/IEF/GLD, where the estimate is -0.0201 (two-sided $p = 0.408$) with an MDE of 0.0679, so its interval includes both a large improvement and none.

6.2.2 A single-regime GARCH model class against static regime covariance

The contrasts in Table 6.2 and Figure 6.1 replace block-constant regime volatilities by GARCH(1,1) volatilities. They compare model classes: the GARCH models have no regime dependence in volatility and also change the correlation input and the mean estimator (Table 4.1). In the pre-declared family, RSC versus D in the four ETF samples, all four estimates are negative. After Holm's adjustment over the four, only the risk-asset sample is significant

Table 6.1: Regime-specific volatility against a pooled normal (B versus A)

Sample	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE
B versus A							
SPY/IEF/GLD	2,516	−0.0201	0.0242	[−0.0675, 0.0274]	−0.83	0.408	0.0679
SPY/IEF/HYG	1,944	−0.1659	0.0373	[−0.2390, −0.0929]	−4.45	<0.001	0.1044
Sectors	2,516	−0.1806	0.0252	[−0.2300, −0.1312]	−7.17	<0.001	0.0705
Risk assets	1,944	−0.1586	0.0217	[−0.2010, −0.1161]	−7.32	<0.001	0.0607
Funds 1992–2019	7,053	−0.1057	0.0293	[−0.1630, −0.0483]	−3.61	<0.001	0.0820

Note: Mean FZ0 difference $\bar{d}$ of the richer model minus the baseline over the forecast days, tail level $\alpha = 5\%$, Gaussian chain, pooled mean; negative favours the richer model. SE: Bartlett-kernel HAC standard error. The interval is $\bar{d} \pm 1.96\,\text{SE}$. The p-value is two-sided. MDE $= 2.8016\,\text{SE}$ is the smallest true difference a two-sided 5% test detects with 80% power. Source: `output/explore/ladder/ladder_dm_table.csv`. Forecast days 2010–2019 (ETFs; credit and risk-asset samples start in 2012), 1992–2019 (funds).

Table 6.2: GARCH margins against static regime covariance

Sample	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE	Holm p
CCC versus B (non-nested)								
SPY/IEF/GLD	2,516	−0.0247	0.0173	[−0.0586, 0.0093]	−1.43	0.154	0.0485	
SPY/IEF/HYG	1,944	−0.0403	0.0269	[−0.0931, 0.0125]	−1.49	0.135	0.0755	
Sectors	2,516	−0.0466	0.0235	[−0.0926, −0.0006]	−1.99	0.047	0.0658	
Risk assets	1,944	−0.0803	0.0230	[−0.1254, −0.0351]	−3.49	<0.001	0.0645	
Funds 1992–2019	7,053	−0.0315	0.0159	[−0.0627, −0.0003]	−1.98	0.048	0.0446	
RSC versus D (non-nested; pre-declared family)								
SPY/IEF/GLD	2,516	−0.0259	0.0173	[−0.0599, 0.0081]	−1.49	0.135	0.0486	0.2341
SPY/IEF/HYG	1,944	−0.0406	0.0259	[−0.0914, 0.0102]	−1.57	0.117	0.0726	0.2341
Sectors	2,516	−0.0396	0.0223	[−0.0832, 0.0040]	−1.78	0.075	0.0623	0.2247
Risk assets	1,944	−0.0760	0.0222	[−0.1195, −0.0325]	−3.43	<0.001	0.0622	**0.0025**
Funds 1992–2019	7,053	−0.0306	0.0198	[−0.0693, 0.0081]	−1.55	0.122	0.0554	–

Note: As Table 6.1. Holm p: the pre-declared family is RSC versus D in the four ETF samples, Holm-adjusted over the four; the fund sample is not in the family. Bold: the one adjusted p-value below 0.05.

(adjusted $p = 0.0025$), so the gate, which required two, was not met. The GARCH advantage is therefore suggestive, not established.

The size of the contrast depends on the mean treatment, though not its sign. With a zero mean in both models the RSC-versus-D estimates range from −0.0339 to −0.0814, larger in absolute value than with the pooled mean; with regime-specific means they are close to the pooled ones (−0.0263 in the main sample; Appendix B).

Calibration points the same way (Table 6.3). On the ETF samples the static models overstate risk, with 5% VaR breached on only 2.8% to 4.2% of days for A, B and D, while the GARCH-margin models breach it on 4.3% to 5.5%. In the fund sample the static models are close to nominal (B 5.4%, D 5.3%).

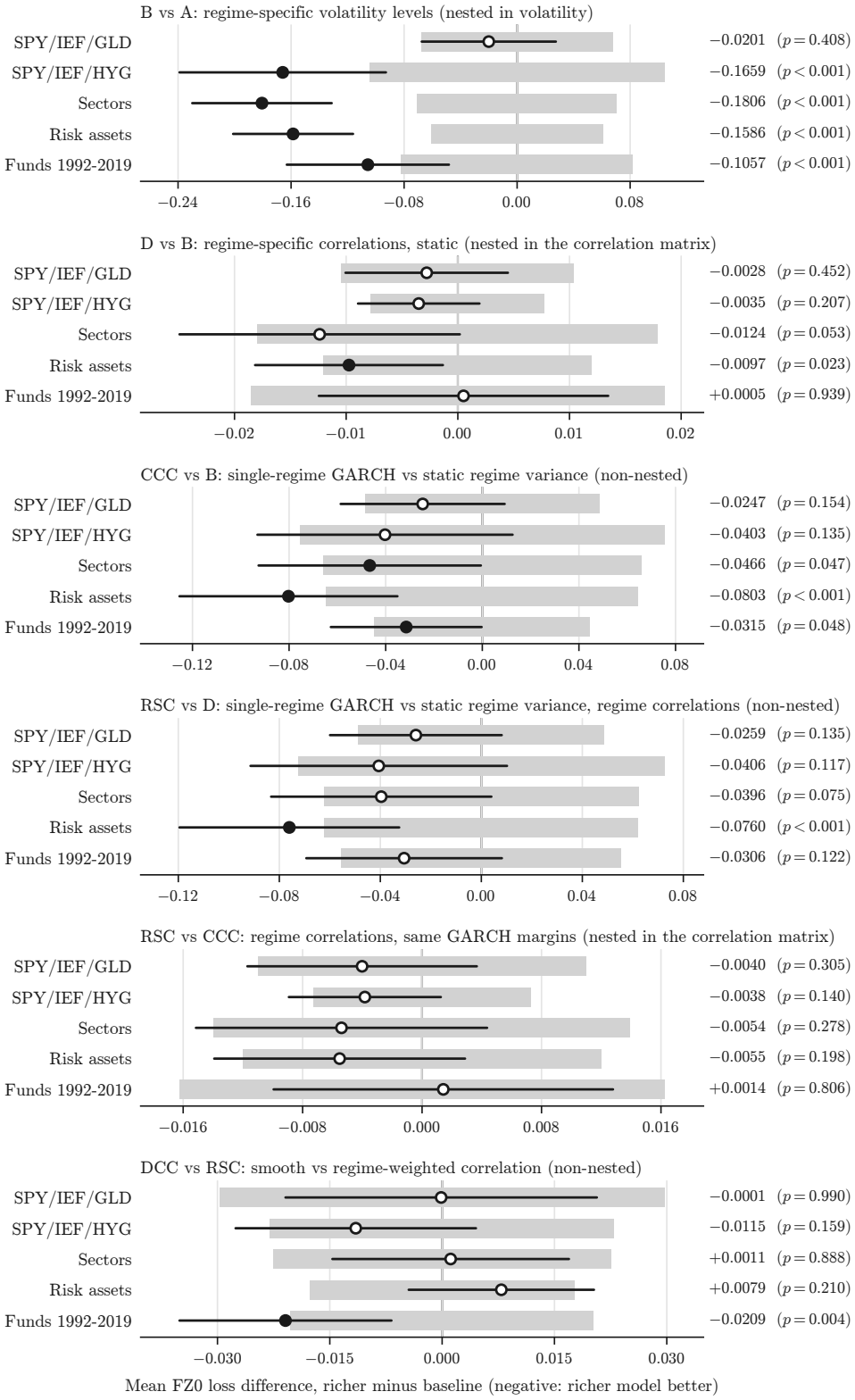


Figure 6.1: Mean FZ0 difference, richer minus baseline, for each comparison and sample

Note: Points are $\bar{d}$; lines are two-sided 95% intervals; the grey band is ± the MDE; filled points have a two-sided $p < 0.05$ (unadjusted). Negative favours the richer model. Each panel has its own horizontal scale: the volatility and margin contrasts are roughly ten times larger than the correlation contrasts. Tail level 5%, Gaussian chain, pooled mean. Source: `output/explore/ladder/ladder_dm_table.csv`, drawn by `scripts/thesis_fig_forest.py`.

Table 6.3: Share of days on which the return fell below the forecast VaR (nominal 5%), development data

Sample	A	B	D	CCC	RSC	DCC
SPY/IEF/GLD	3.1%	3.7%	3.5%	5.0%	4.7%	5.0%
SPY/IEF/HYG	3.4%	4.2%	4.0%	5.5%	5.5%	5.0%
Sectors	3.1%	3.4%	3.7%	5.2%	5.4%	5.5%
Risk assets	2.8%	2.8%	3.0%	4.3%	4.6%	4.7%
Funds 1992–2019	4.8%	5.4%	5.3%	5.5%	5.2%	5.5%

Note: Gaussian chain, pooled mean, forecast days as in Table 6.1. A hit is a day with $y_t \leq \mathrm{VaR}_t$. Source: `output/explore/ladder/ladder_loss_table.csv`.

6.2.3 The correlation steps

Letting correlations switch, with the volatility model held fixed, never improved the loss detectably (Table 6.4). With GARCH margins, switching all correlations (RSC versus CCC) gives estimates between -0.0040 and -0.0055 in the four ETF samples and $+0.0014$ in the funds; every interval includes zero (two-sided p-values from 0.140 to 0.806), and the estimates are between about a third and a half of their MDEs (0.36 to 0.52). The data are consistent with gains up to about the MDE. The answer is not the same under every mean treatment: with a zero mean the sector sample gives -0.0087 (two-sided $p = 0.046$, unadjusted) and the risk-asset sample -0.0070 ($p = 0.080$). The smooth alternative, DCC, is indistinguishable from RSC in the ETF samples (p from 0.159 to 0.990) and better than RSC in the fund sample by 0.0209 ($p = 0.004$, unadjusted).

The static step D versus B needs a caution. Under the simulated process with GARCH margins and no change in correlation (scenario S0), the one-sided test of D against B rejects in 9.6% of replications (95% interval 7.0% to 12.2%) at 1,508 test days, in 11.0% (6.7% to 15.3%) at 2,520 days, and in 14.8% and 17.0% of replications at $|t| > 1.96$ (Table 6.6). The test is oversized when volatility clusters. In the exploration grid, 37 of the 214 headline D-versus-B tests have a raw $p < 0.05$ (17.3%), 18 favouring D and 19 favouring B, and none survives a Holm adjustment (smallest adjusted $p = 0.1413$). That rate matches the simulated null rate, so the raw count is not evidence of an effect. The simulated process is stylised and the 214 tests are

Table 6.4: Correlation steps

Sample	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE
D versus B (nested in the correlation matrix; size unreliable under GARCH margins)							
SPY/IEF/GLD	2,516	−0.0028	0.0037	[−0.0101, 0.0045]	−0.75	0.452	0.0104
SPY/IEF/HYG	1,944	−0.0035	0.0028	[−0.0089, 0.0019]	−1.26	0.207	0.0078
Sectors	2,516	−0.0124	0.0064	[−0.0249, 0.0002]	−1.93	0.053	0.0179
Risk assets	1,944	−0.0097	0.0043	[−0.0182, −0.0013]	−2.27	0.023	0.0120
Funds 1992–2019	7,053	0.0005	0.0066	[−0.0125, 0.0135]	0.08	0.939	0.0185
RSC versus CCC (nested in the correlation matrix)							
SPY/IEF/GLD	2,516	−0.0040	0.0039	[−0.0117, 0.0037]	−1.02	0.305	0.0110
SPY/IEF/HYG	1,944	−0.0038	0.0026	[−0.0089, 0.0013]	−1.48	0.140	0.0073
Sectors	2,516	−0.0054	0.0050	[−0.0151, 0.0044]	−1.08	0.278	0.0139
Risk assets	1,944	−0.0055	0.0043	[−0.0139, 0.0029]	−1.29	0.198	0.0120
Funds 1992–2019	7,053	0.0014	0.0058	[−0.0099, 0.0128]	0.25	0.806	0.0162
DCC versus RSC (non-nested)							
SPY/IEF/GLD	2,516	−0.0001	0.0106	[−0.0209, 0.0207]	−0.01	0.990	0.0297
SPY/IEF/HYG	1,944	−0.0115	0.0082	[−0.0276, 0.0045]	−1.41	0.159	0.0229
Sectors	2,516	0.0011	0.0081	[−0.0147, 0.0169]	0.14	0.888	0.0226
Risk assets	1,944	0.0079	0.0063	[−0.0045, 0.0203]	1.25	0.210	0.0177
Funds 1992–2019	7,053	−0.0209	0.0072	[−0.0351, −0.0068]	−2.90	0.004	0.0202

Note: As Table 6.1.

not independent, so the match is suggestive, not proof.

6.3 The primary pair on the development data

On the development data the confirmatory comparison, RSC_P against CCC, gives an estimate of −0.0015 over 2010 to 2019 (two-sided $p = 0.136$, MDE 0.0028), −0.0012 in 2010–2015 ($p = 0.371$) and −0.0019 in the dry-run years 2016–2019 ($p = 0.198$; Table 6.5). These estimates are about the size of the smallest effect of interest and not significant. Their standard errors, 0.0010, 0.0013 and 0.0015, are the ones relevant to the hold-out, which has 1,508 days. Because the pair, the model and the test were chosen with these days in view, they validate nothing.

6.4 Power at the hold-out length

At 1,508 test days the primary test is correctly sized and has low power (Table 6.6). Its size, the rejection rate in S0, is 3.6%, with no sign of oversizing, although 500 replications cannot show that it is at most 5%; the mean difference in S0 is slightly positive, so the null is interior rather than on the boundary. The power is 18.2% for a moderate shift (S1), 35.2% for a strong one

Table 6.5: Single-pair correlation switching against constant correlation, development data (RSC_P versus CCC)

α	Forecast days	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE
5.0%	2010–2019	2,516	−0.0015	0.0010	[−0.0034, 0.0005]	−1.49	0.136	0.0028
5.0%	2010–2015	1,510	−0.0012	0.0013	[−0.0038, 0.0014]	−0.89	0.371	0.0037
5.0%	2016–2019	1,006	−0.0019	0.0015	[−0.0048, 0.0010]	−1.29	0.198	0.0041
2.5%	2010–2019	2,516	−0.0028	0.0015	[−0.0058, 0.0002]	−1.82	0.069	0.0043
2.5%	2010–2015	1,510	−0.0029	0.0021	[−0.0070, 0.0012]	−1.37	0.171	0.0059
2.5%	2016–2019	1,006	−0.0027	0.0022	[−0.0070, 0.0017]	−1.20	0.231	0.0062

Note: RSC_P minus CCC, Gaussian chain, pooled mean; negative favours switching the SPY–IEF correlation. SE is MDE/2.8 (the table stores MDE rounded to 2.8 SE); the interval is $\bar{d} \pm 1.96\,\mathrm{SE}$. These are development data: the specification was chosen after seeing them, so they are not validation. Source: `output/explore/routes/routes_dm_table.csv`, provider HMM_SPY2.

(S2) and 2.4% when correlations move the other way (S3), against 27.0% and 53.5% at 2,520 test days. The pre-declared low-power rule, triggered if S2 power fell below 30%, therefore did not apply, but a moderate shift would be detected about one time in five.

Simulated power does not transfer one-for-one to the real data. The mean standard error of the primary difference in the simulation depends on the scenario: it is 0.0006 in S0, 0.0010 in S1 and 0.0026 in S2, because a larger shift in correlation makes the loss difference more variable. The real-data standard errors above, 0.0010 to 0.0015, lie between the S0 and S2 values, so the power of 35.2% in S2 reflects both a mean difference of −0.0034 and a standard error of 0.0026, and it is not the power that the same difference would have on real data. The minimum detectable difference at the real standard error, about $2.4865 \times 0.0014 = 0.0035$ for the one-sided test (0.0031 at the standard error observed on the hold-out, Chapter 7), is the more transferable summary.

Table 6.6: Simulated size and power of the difference tests (500 replications per scenario at 1,508 days; 200 at 2,520 days)

Scenario	One-sided rejection [95% CI]	Two-sided	Upper tail	Mean $\bar{d}$	Mean SE
Primary test: RSC_P versus CCC, 1,508 test days, 500 replications					
S0 (no correlation change)	3.6% [2.0%, 5.2%]	7.4%	5.2%	0.0002	0.0006
S1 (moderate)	18.2% [14.8%, 21.6%]	14.2%	0.4%	−0.0009	0.0010
S2 (strong)	35.2% [31.0%, 39.4%]	24.2%	0.0%	−0.0034	0.0026
S3 (flight to quality)	2.4% [1.1%, 3.7%]	4.8%	3.4%	0.0002	0.0017
Same test, 2,520 test days (earlier study, 200 replications)					
S0 (no correlation change)	4.5% [1.6%, 7.4%]	7.0%	4.5%	0.0001	0.0005
S1 (moderate)	27.0% [20.8%, 33.2%]	17.5%	0.0%	−0.0009	0.0008
S2 (strong)	53.5% [46.6%, 60.4%]	44.0%	0.0%	−0.0035	0.0020
S3 (flight to quality)	5.5% [2.3%, 8.7%]	3.5%	1.0%	0.0001	0.0013
RSC versus CCC (all pairs switch), 1,508 test days					
S0 (no correlation change)	4.6% [2.8%, 6.4%]	7.2%	4.2%	0.0004	0.0018
S1 (moderate)	17.2% [13.9%, 20.5%]	9.8%	0.4%	−0.0022	0.0032
S2 (strong)	25.4% [21.6%, 29.2%]	17.0%	0.0%	−0.0069	0.0069
S3 (flight to quality)	2.6% [1.2%, 4.0%]	6.6%	5.2%	0.0010	0.0036
Static D versus B, 1,508 test days (size under GARCH margins)					
S0 (no correlation change)	9.6% [7.0%, 12.2%]	14.8%	8.0%	0.0002	0.0024

Note: Tail level $\alpha = 5\%$, Gaussian chain, simulated data from the stylised GARCH regime-switching process (Section 5.3.2); S0 is the null and its row is the test size. One-sided: reject when $t < -1.6449$ (the pre-declared rule). Two-sided: $|t| > 1.96$. Upper tail: $t > +1.96$, the rate at which the richer model is declared significantly worse. The interval is a 95% binomial interval on the replications. Mean $\bar{d}$ and mean SE are averages over replications. Sources: `power_1508_summary.csv`, `power_2520_twosided.csv`.

Chapter 7: Results on the Hold-Out

This chapter reports the single pre-specified test on the hold-out (1,508 forecast days) with its frozen outcome sentence (Section 5.2.3), then the secondary analyses, calibration and post hoc checks. The forecasts were produced once from the code fixed by the tag `spec-freeze-v1`; the run was clean, and the scorer's provenance check was amended once before any number was computed (Appendix A).

7.1 The pre-specified test

Letting the SPY–IEF correlation switch with the regime (RSC_P) lowered the average loss against constant correlation (CCC) by 0.0013, a difference the test could not distinguish from zero (Table 7.1). With a HAC standard error of 0.0013, $t = -1.00$ and the pre-declared one-sided p-value is 0.158, above the 5% level (the critical value is -1.6449). The two-sided 95% interval runs from -0.0037 to $+0.0012$, and the statistic is far from the $+1.96$ that would mark the switching model as significantly worse.

This is outcome O2, because the one-sided lower bound, -0.0033, is below the smallest effect of interest (-0.0020), so O3 is not reached. The frozen sentence is:

> The hold-out data neither establish an improvement nor exclude one. The estimate is -0.0013 (95% CI $[-0.0037, +0.0012]$); an improvement of at least 0.0031 (the minimum detectable difference for the one-sided 5% test, 80% power) would have been detected with 80% probability, and smaller improvements, including none, are consistent with the data.

The frozen sentence needs two qualifications. First, the estimate is 0.4 times the minimum detectable difference, so the design could not detect an improvement of this size with any useful probability; the standard error, 0.0013, lies within the range of 0.0010 to 0.0015 that the development data had suggested (Section 6.3). Second, at the standard error observed, a true

Table 7.1: The pre-specified hold-out test: switching SPY–IEF correlation against constant correlation

Quantity	Value
Forecast days n	1,508
Mean difference $\bar{d}$ (RSC_P minus CCC)	−0.0013
HAC standard error (7 lags)	0.0013
t statistic	−1.00
One-sided p-value (pre-declared test)	0.158
Two-sided p-value	0.316
95% interval, two-sided	[−0.0037, 0.0012]
One-sided 95% lower bound $\bar{d} - 1.6449\,\mathrm{SE}$	−0.0033
Smallest effect of interest (SESOI)	−0.0020
MDE, one-sided 5% test, 80% power	0.0031
MDE, two-sided 5% test, 80% power	0.0035
Simulated power, moderate shift S1 / strong shift S2	18.2% / 35.2%
Outcome	O2

Note: Forecast days 2 January 2020 to 31 December 2025, tail level $\alpha = 5\%$, Gaussian chain, pooled mean. Negative $\bar{d}$ favours switching the SPY–IEF correlation. The interval is $\bar{d} \pm 1.96\,\mathrm{SE}$. Power is the simulated one-sided rejection rate at 1,508 days (Table 6.6), not a property of this estimate. Source: `output/confirm/holdout/primary.json`.

improvement equal to the smallest effect of interest, 0.0020, would have been detected with a probability of about 48%; this figure was computed after the result was known, for that stated effect and not at the estimate. I therefore claim neither equivalence with constant correlation nor an improvement.

7.2 Secondary comparisons

These are the secondary analyses declared in Section 5.5; they are unadjusted, not confirmatory, and leave the outcome unchanged.

The ladder on the hold-out (first panel of Table 7.2) repeats the development pattern without adding precision. Switching all correlations gives −0.0037 ($p = 0.265$, MDE 0.0092). The three comparisons that involve the pooled-normal and static models have estimates between −0.095 and −0.130 but standard errors more than twenty times those of the primary pair, so the hold-out says very little about them. The static step D against B is positive, +0.0155 ($p = 0.186$), and carries the oversizing caution of Chapter 6.

Changing only the tail level or the chain leaves the sign unchanged: at $\alpha = 2.5\%$ the estimate

Table 7.2: Secondary comparisons on the hold-out (unadjusted; not confirmatory)

Comparison	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE
Ladder steps (all pairs switch where stated)							
B versus A	1,508	−0.0952	0.0879	[−0.2675, 0.0771]	−1.08	0.279	0.2463
D versus B	1,508	0.0155	0.0117	[−0.0074, 0.0385]	1.32	0.186	0.0328
CCC versus B	1,508	−0.1107	0.0820	[−0.2715, 0.0501]	−1.35	0.177	0.2298
RSC versus D	1,508	−0.1299	0.0919	[−0.3099, 0.0502]	−1.41	0.157	0.2574
RSC versus CCC	1,508	−0.0037	0.0033	[−0.0101, 0.0028]	−1.11	0.265	0.0092
DCC versus RSC	1,508	−0.0060	0.0148	[−0.0351, 0.0230]	−0.41	0.683	0.0415
Primary pair at other settings							
α 2.5%, Gaussian chain	1,508	−0.0026	0.0022	[−0.0070, 0.0017]	−1.19	0.233	0.0062
α 5%, Student-t chain	1,508	−0.0012	0.0012	[−0.0035, 0.0012]	−0.96	0.339	0.0034
α 2.5%, Student-t chain	1,508	−0.0023	0.0020	[−0.0062, 0.0016]	−1.16	0.245	0.0055
Calendar year 2022 (interval only, not a test)							
RSC_P versus CCC, 2022	251	−0.0009	0.0041	[−0.0089, 0.0070]	–	–	–

Note: Mean FZ0 difference of the richer model minus the baseline; negative favours the richer model. Panel 1 is the ladder of Chapter 4 on the hold-out, with tail level 5%, Gaussian chain, pooled mean. The interval is $\bar{d} \pm 1.96\,\mathrm{SE}$ and the MDE is $2.8016\,\mathrm{SE}$ (two-sided 5%, 80% power). None of these p-values is adjusted and none is cited as confirmation. Source: `output/confirm/holdout/secondary_ladder.csv`, `secondary_alpha_and_t_chain.csv`, `secondary_focus_year.csv`.

is −0.0026 ($p = 0.233$), and with a Student-t chain at $\alpha = 5\%$ it is −0.0012 ($p = 0.339$). For the calendar year 2022 alone, the year of the reversal, the estimate is −0.0009 over 251 days, with an interval from −0.0089 to +0.0070, too wide to say anything about the year.

The Giacomini–White conditional test asks whether the loss difference depends on the lagged 63-day SPY–IEF correlation, the economic prediction of Chapter 1. It finds no such dependence (Wald statistic 1.40 on two degrees of freedom, $p = 0.498$). In the data, 667 of 1,508 days (44.2%) had a positive lagged correlation; the mean loss difference was −0.0004 on those days and −0.0019 on the others. These means are descriptive, and their direction runs opposite to the mechanism that motivated the question. The test of superior predictive ability against GARCH-t, with the candidates B, D, RSC and RSC_P, is also non-significant (consistent $p = 0.477$).

7.3 Calibration

Both models were breached more often than intended, by about the same amount (Table 7.3). At the 5% level the VaR was breached on 6.9% of days by RSC_P and 6.8% by CCC (104 and 103 days, against 75 expected), and the Kupiec test rejects correct coverage for both ($p = 0.001$ and 0.002). At 2.5% the breach rates were 4.4% and 4.3% (Kupiec $p < 0.001$). The independence

Table 7.3: Coverage and expected-shortfall calibration of the two primary models on the hold-out

α	Model	Hits	Hit rate	Kupiec p	Indep. p	Cond. cov. p	Z_2	Z_2 p	ES ratio
Gaussian chain (the primary forecasts)									
5.0%	RSC_P	104	6.9%	0.001	0.480	0.005	0.616	<0.001	1.17
5.0%	CCC	103	6.8%	0.002	0.444	0.006	0.606	<0.001	1.18
2.5%	RSC_P	66	4.4%	<0.001	0.947	<0.001	1.083	<0.001	1.19
2.5%	CCC	65	4.3%	<0.001	0.903	<0.001	1.064	<0.001	1.20
Student-t chain (POST HOC, descriptive)									
5.0%	RSC_P	108	7.2%	<0.001	0.399	<0.001	–	–	1.08
5.0%	CCC	108	7.2%	<0.001	0.399	<0.001	–	–	1.08
2.5%	RSC_P	63	4.2%	<0.001	0.817	<0.001	–	–	1.07
2.5%	CCC	63	4.2%	<0.001	0.817	<0.001	–	–	1.07

Note: 1,508 forecast days; 75.4 hits expected at 5% and 37.7 at 2.5%. A hit is a day with $y_t \leq \mathrm{VaR}_t$. Kupiec: unconditional coverage; Indep.: Christoffersen independence; Cond. cov.: their sum. Z_2 is positive when risk is underestimated (my sign, opposite to Acerbi and Szekely); its p-value is the simulated share of Z_2 values at or above the observed one, and it is not simulated for the Student-t chain. ES ratio: mean of y_t/ES_t over hit days, above one when exceedances are larger than the forecast shortfall. Descriptive, not part of the decision. Sources: `output/confirm/holdout/calibration.csv`; Student-t rows `output/confirm/posthoc.json` (`scripts/thesis_posthoc.py`).

test finds no clustering ($p = 0.480$ and 0.444), though with about a hundred breaches it has little power.

Breach-day losses were also larger than forecast. Z_2 is positive for both models (0.62 and 0.61) with a simulated p-value of 0.0002, the smallest that 5,000 draws can give. Because Z_2 combines the frequency of breaches with their size, the size alone is better read from the average return on breach days, about 1.17 to 1.18 times the forecast shortfall. In the development data the GARCH-margin models had been close to nominal (Table 6.3), so the shortfall is specific to 2020–2025. Because both models fell short together, their near-equal losses say nothing about whether either is fit for use.

7.4 Post hoc checks

The checks in this section were chosen after the result was known. They are descriptive, change no outcome, and were computed from the stored forecasts without refitting anything.

7.4.1 Where the difference arises

Figure 7.1 plots the running sum of the daily loss differences. The total over all 1,508 days is −1.91. The switching model lost ground in the 2020 crash: the days from 15 February to 30

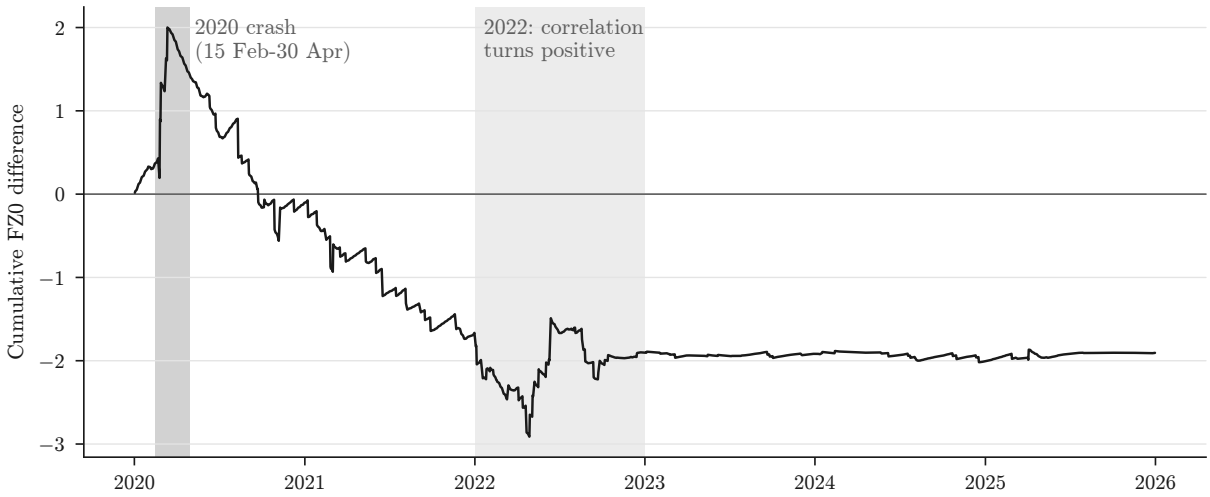


Figure 7.1: Cumulative daily FZ0 loss difference on the hold-out, switching minus constant correlation

Note: Running sum of $d_t = \mathrm{FZ0}_t^{\mathrm{RSC_P}} - \mathrm{FZ0}_t^{\mathrm{CCC}}$, tail level 5%, Gaussian chain; a falling line means the switching model had the lower loss. The darker band is the crash window of 15 February to 30 April 2020, the lighter band the calendar year 2022. Source: `output/confirm/holdout_forecasts.pkl`, drawn by `scripts/thesis_fig_cumulative.py`.

April 2020 sum to +1.05, and excluding that window the mean difference is −0.0020. It then gained steadily through 2021, which contributes −1.56, against −0.10 for 2020 as a whole and −0.24 for 2022, and the two models tracked each other from 2023. The average is therefore the net of a few large, opposite-signed episodes, with 2021, when the calendar-year SPY–IEF correlation was −0.09, contributing most of the favourable part and not 2022.

7.4.2 Inference

The test used a HAC lag of 7. With no lag the standard error is 0.0012 ($t = -1.04$), with 21 lags 0.0014 ($t = -0.90$) and with 42 lags 0.0015 ($t = -0.84$). Because the loss difference is dominated by a few days, I also computed stationary-bootstrap intervals for the mean, which do not rely on the normal approximation: with expected block lengths of 5, 21 and 63 days they run from −0.0037 to +0.0013, from −0.0039 to +0.0017 and from −0.0042 to +0.0017. Every choice leaves the result non-significant, and the normal interval is about as wide as the bootstrap ones.

Table 7.4: Share of hold-out days on which the 5% VaR was breached, by calendar year

	2020	2021	**2022**	2023	2024	2025
Forecast days	253	252	**251**	250	252	250
CCC (constant correlation)	6.3%	7.1%	**12.0%**	6.4%	6.0%	3.2%
RSC_P (switching SPY–IEF correlation)	6.3%	7.1%	**12.4%**	6.4%	6.0%	3.2%
RiskMetrics EWMA (POST HOC baseline)	6.7%	6.0%	**6.0%**	4.0%	5.2%	3.6%

Note: Nominal rate 5%; Gaussian chain for CCC and RSC_P. 2022, the year in which the stock–bond correlation turned positive, is in bold. POST HOC and descriptive: chosen after the result was known; no test. Source: `output/confirm/posthoc.json` (`scripts/thesis_posthoc.py`).

7.4.3 Baselines

Three simpler forecasts of the portfolio return, scored with the same loss on the same days, were statistically indistinguishable from constant correlation. The mean FZ0 difference against CCC (positive means worse than CCC) was +0.0317 for GARCH-t (HAC standard error 0.0429), +0.1308 for historical simulation (0.1278) and +0.0063 for the RiskMetrics EWMA (0.0440), and every interval contains zero. Their 5% breach rates were 6.6%, 5.8% and 5.2%, against 6.8% and 6.9% for CCC and RSC_P; the Kupiec test rejects correct coverage for GARCH-t ($p = 0.008$) but not for historical simulation ($p = 0.146$) or the EWMA ($p = 0.673$). A one-parameter EWMA therefore came close to nominal coverage with a loss indistinguishable from that of the GARCH-based models.

7.4.4 Breaches by year

The excess breaches of the two primary models were concentrated in one year (Table 7.4). In 2022, constant correlation breached its 5% VaR on 12.0% of days and the switching model on 12.4%, more than twice the intended rate, while the EWMA breached on 6.0%. In the other years the two primary models stayed between 3.2% and 7.1%.

7.4.5 Heavier tails

The Student-t versions of the two primary models (lower panel of Table 7.3) breached the 5% VaR more often, on 7.2% of days, and the 2.5% VaR on 4.2%, while breach-day losses came closer to the forecast shortfall (ratio 1.07 to 1.08). Fatter tails did not fix the breach rate, which

is consistent with forecast volatility that was too low rather than tails that were too thin.

Chapter 8: Contributions, Limitations and Future Work

8.1 Contributions

The thesis makes three contributions. First, it evaluates regime-switching correlation for a stock–bond–gold portfolio with a joint VaR–ES loss, which the closest studies do not: Paolella et al. (2019) use density and coverage scores on stocks, and Fortin et al. (2023) use no regime switching. Second, it separates the volatility, correlation and mean channels under one regime signal. On the development data the gains come from modelling volatility, through regime-specific or GARCH volatility, and not from switching correlations, whose estimates were a third to a half of what the design could detect. Third, it reports a hold-out result with its interval, minimum detectable difference and power fixed in advance, so that a reader can see which effect sizes the data rule out and which they leave open.

The hold-out result is neither for nor against regime-switching correlation (Section 7.1), and no difference can be attributed to a single channel where comparisons change several things (Table 4.1). The decomposition is the part I would defend most: for each step it gives an estimate, an interval and what the design could detect.

Methodologically, the thesis shows that a common shortcut fails: the static test of regime-specific against pooled correlation is oversized when volatility clusters (Section 6.2.3), so raw rejection counts across many specifications can look like evidence where there is none.

8.2 What a risk desk should take from the hold-out

For a practitioner the calibration result matters most: both models under-forecast risk by the same margin (Section 7.3), so the choice between them is secondary. The shortfall was concentrated in 2022, when both breached their 5% VaR on about 12% of days against 6.0% for the EWMA (Table 7.4). The evidence points to the level of forecast volatility rather than to the correlation model or the tail shape. The portfolio's volatility rose from 7.1% a year in the development period to 10.0% in the hold-out (Table 3.1); the Student-t versions breached more

often; and the EWMA, whose variance adapts within weeks, held close to nominal coverage (Section 7.4). GARCH margins fitted on a five-year window and refitted every 63 days registered the higher risk more slowly. This explanation fits the evidence but is untested, and the EWMA comparison was chosen after the result was known. For a portfolio like this one, the practical advice is to check the scale of the forecast against recent volatility before choosing a correlation model; the evidence gives no reason to adopt regime-switching correlation.

8.3 Reading the result against the economic framework

The economic prediction was that a signal built from equity volatility should help most when the recent stock–bond correlation is positive. The hold-out runs the other way: the gain did not depend on the lagged correlation, came mostly in 2021, and reversed in the 2020 crash (Sections 7.2 and 7.4). This is descriptive, but it fits better with the macroeconomic account of Campbell et al. (2020) and Molenaar et al. (2024), in which the sign of the correlation follows inflation and real rates rather than equity volatility.

Figure 8.1 shows where the mechanism broke. The regime signal did register 2022 as stressed: the predicted stress probability exceeded one half on 56% of its days, against 41% in 2020. But the stress-state SPY–IEF correlation that the model had learned was between -0.40 and -0.16 during 2022, more negative than its calm-state estimate (-0.25 to -0.20). The stress days in its five-year window, such as the 2020 crash, had been flight-to-quality days on which bonds rose as equities fell, and positive correlation had been rare before 2022 (Table 3.2). Switching to the stress state therefore made the forecasts assume more hedging just as the realised correlation rose to about $+0.5$, and the stress-state estimate first turned positive only in March 2025. The difficulty lay less in the speed of the signal than in what stress had meant in the data the model learned from: a window refitted every 21 days could not learn a positive stress correlation it had not yet seen. This reading is descriptive, and no variant was run on the hold-out because none was pre-specified.

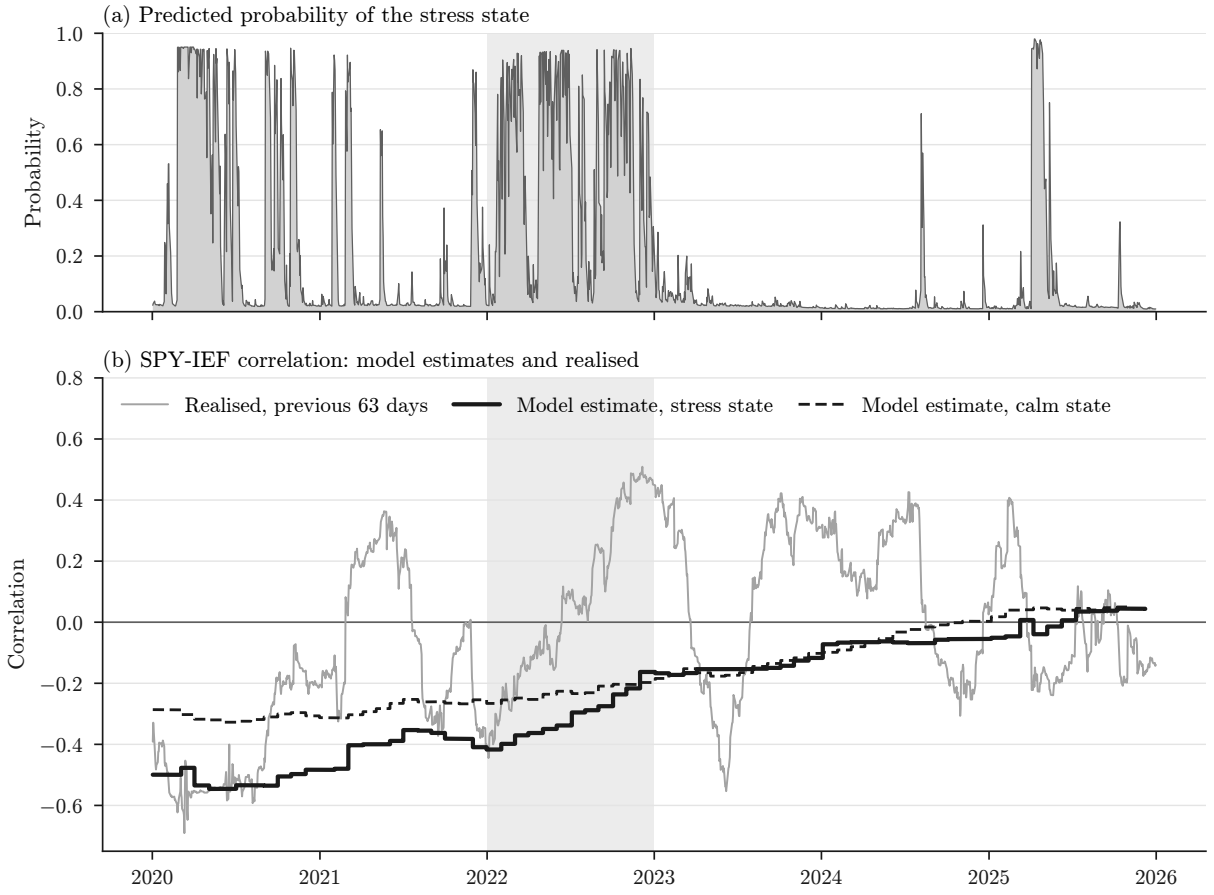


Figure 8.1: What the switching model believed on the hold-out: stress probability and SPY–IEF correlation estimates

Note: (a) Predicted probability of the stress state for each forecast day. (b) The switching model's calm-state and stress-state SPY–IEF correlation estimates (standardised residuals, re-estimated every 21 days) against the realised correlation of SPY and IEF returns over the previous 63 days. The shaded band is 2022. Descriptive, from the stored hold-out forecasts; nothing is refitted. Source: `output/confirm/holdout_forecasts.pkl`, drawn by `scripts/thesis_fig_regime.py`; summary in `output/confirm/regime_summary.json`.

8.4 Limitations

The following limits apply to every result in the thesis.

- **Power.** The test could detect only differences above 0.0031, more than the smallest effect of interest of 0.0020 (Section 5.3.3). It measures how precisely this sample can speak rather than offering a realistic chance of excluding a meaningful gain, and a longer ETF history would not add the regime of interest (Table 3.2).

- **Selection.** The comparison was chosen after exploring ten universes and ten regime definitions, from a simulation in which the SPY–IEF shift is largest by construction. Nothing

was fitted on the hold-out, but the specification depends on the exploration, and the pre-registration is self-registered without an independent timestamp.

- **A stylised power study.** The simulated process is assumed, and its standard errors differ from the real ones (Section 6.4), so its power figures need not transfer to markets; its size of 3.6% and upper-tail rate of 8.0% at 1.645 are rough checks of the normal reference.
- **Inference.** The estimand concerns forecasting procedures, not population parameters (Section 5.2.1); the uncertainty of the estimated regimes is not carried into the test; and a few episodes in 2020 and 2021 dominate the average.
- **Baselines.** The practitioner baselines, historical simulation and EWMA, were added after the hold-out was scored. They should have been part of the frozen design.
- **Signal and scope.** One equity-volatility signal with a five-year window; one portfolio of three funds with constant weights, one data vendor and one reversal episode; and a 5% tail level chosen for power, whereas regulatory backtests use the 1% VaR. I make no claim about other portfolios, tail levels or markets.

8.5 Future work

Each extension below would be a new study with its own pre-specification. A regime signal built from the stock–bond correlation itself or from inflation and real rates would test directly whether a slow macroeconomic state helps where a volatility state did not. A window that adapts to the regime, or exponentially weighted estimates, would let the model learn a new stress relationship sooner, and the EWMA result suggests that forecast scale should be part of any such comparison. A broader sample, such as several stock–bond portfolios or international funds, would raise power, and a power calculation like the one in Chapter 5 should come first. A bootstrap of the whole estimation pipeline would carry the uncertainty of the regime estimates into the test.

Chapter 9: Conclusion

I asked whether letting the stock–bond correlation switch with a market regime improves one-day value-at-risk and expected-shortfall forecasts of an equal-weight portfolio of SPY, IEF and GLD. I built a ladder of models that separates the treatment of volatility, correlation and the mean under one regime signal, fixed a single confirmatory test before using 2020–2025, and ran it once.

The answer is that six years of daily data cannot tell a useful gain from none: switching lowered the average loss slightly, the pre-specified test did not reject (outcome O2), and only gains above 0.0031 were detectable. Regime information helped through volatility rather than correlation, and in 2022 the model expected more hedging just as the hedge failed (Section 8.3).

For a risk manager of a portfolio like this one, the evidence gives no reason to adopt the more complex model, though it leaves open that the model helps. The more immediate problem is calibration: both models under-forecast risk over 2020–2025, while a simple EWMA, scored post hoc, came close to nominal coverage with a loss indistinguishable from theirs. Checking the scale of the forecast against recent volatility matters more here than the choice of correlation model.

These conclusions rest on one portfolio, one vendor, one reversal episode, a regime signal built from equity returns and a power study that rests on a stylised process (Section 8.4). Whether a regime signal tied to inflation, or a larger sample, would show the gain that this one could not is a question for a new, separately pre-specified study.

Chapter 10: Code and Data

This chapter explains how the analysis is organised in the accompanying repository, how the data are stored and transformed, and how every table and figure can be reproduced. Paths are relative to the repository root. The code is Python 3.9.6; package versions are pinned in `requirements.txt`, and the main ones are numpy 2.0.2, pandas 2.3.3, scipy 1.13.1, arch 7.2.0, statsmodels 0.14.6, hmmlearn 0.3.3 and matplotlib 3.9.4.

10.1 Structure of the repository

`config/pilot.yaml` The single source of parameters and seeds: tickers, window (1,260 days), refit intervals (21 and 63 days), tail levels and the simulation design. Its pilot-stage lock date (2016) predates the hold-out design and is not used by the confirmatory run.

`data/raw/` The raw Yahoo Finance downloads, never re-pulled, with SHA-256 hashes in `MANIFEST.txt`. The thesis uses `spy_ief_gld_raw_2026-09-26.csv` (main sample) and `explore_etfs_2004_2019.csv` and `explore_rates_1986_2019.csv` (development samples). `data/processed/` and `data/metadata/` hold pilot-stage files that some tests read.

`src/` The models and tests. `returns.py` builds returns and the portfolio; `engine.py` holds the hidden Markov model, the mixture VaR and ES, the FZ0 loss, the Diebold–Mariano test and the coverage tests; `explore_wf.py` runs the walk-forward forecasts of the static models, DCC and GARCH-t; `explore_routes.py` runs the GARCH-margin models CCC, RSC and RSC_P; `explore_ladder.py` assembles the ladder; `explore_inference.py` implements the Giacomini–White test; `dgp_garch.py` simulates the process of the power study.

`scripts/` Entry points, in three groups: `explore_00` to `explore_22` (development results and power studies, Chapter 6); `confirm_00` to `confirm_06` (the frozen hold-out runner and scorer and their checks, Chapter 7); and `thesis_*` together with `build_all.py` and

`check_numbers.py` (tables, figures, post hoc checks and the number trace).

`output/` Results: `explore/` (development), `confirm/` (dry run, hold-out and post hoc), `figures/` and `tables/`.

`docs/` The confirmatory specification (`preregistration_confirmatory.md`) and the log of every specification tried in development (`decisions_log.md`).

`tests/` 101 unit tests run by default, and three more that read hold-out-period data and were run only after scoring.

`thesis/` The LaTeX source (`tex/`), the generated tables (`tex/generated/`), the number registry (`numbers.yaml`) and the compiled thesis (`final/thesis.pdf`).

The files under `src/`, `scripts/confirm_00_run.py` and `requirements.txt` are frozen by the tags `spec-freeze-v1` and `spec-freeze-v1.1`: the scorer refuses to score forecasts produced by code that differs from the tagged version. Pilot-stage modules in `src/` therefore remain in place even though no thesis table depends on them.

10.2 Data storage and transformations

The raw file stores every Yahoo field for the three funds from 18 November 2004 to 25 September 2026. The forecasting code reads only the adjusted closes, truncates them at 31 December 2025, drops dates with a missing price, and computes simple returns between consecutive available dates (`src/returns.py`). The portfolio return is the equal-weighted mean of the three asset returns. Nothing is forward-filled, winsorised or rescaled, except that the GARCH estimation works with returns in per cent and converts back.

Because Yahoo revises adjusted prices when it recomputes distributions, the numbers depend on the data vintage; this is why the raw files are stored with their hashes. A second download of the same three tickers, truncated at 2019, had the same 3,804 dates, and its daily returns differed from the first by at most 2.5×10^{-6} (portfolio returns by at most 9.3×10^{-7}). This

shows that the download was reproducible across two pulls; it is not a second source, since every ETF series comes from one vendor.

10.3 Implementation settings

The hidden Markov model is fitted with `hmmlearn` (diagonal covariance, tolerance 10^{-4}, seed 0). The first fit of a run is a cold start with 100 iterations from start probabilities $(0.5, 0.5)$, a transition matrix with diagonal $(0.98, 0.95)$, both means at the window mean, and variances $0.5v$ and $2.5v$, where v is the window variance; every later fit warm-starts from the previous block's estimates with at most 50 iterations. Because the chain carries its estimates forward, a run that starts in 2016 differs slightly from the exploration chain that started in 2010 (a median absolute VaR difference of 1.3×10^{-5} to 1.4×10^{-5} in the primary models); the hold-out run starts cold in the same way.

In the GARCH-margin models, each window is re-filtered with the fitted parameters, starting from the sample variance of the window's residuals; the variance for the first day of a block continues the recursion, and later days in the block are updated from the realised return with the parameters held fixed. A regime or hybrid correlation matrix whose smallest eigenvalue is below 10^{-8} is repaired by clipping the eigenvalues and rescaling to a unit diagonal. The repair never fired: 0 of 240 matrices in the chain that starts in 2010, 0 of 96 in the chain that starts in 2016, and 0 of 144 in the hold-out run. If a DCC fit returns $\theta_1 < 0$, $\theta_2 < 0$ or $\theta_1 + \theta_2 \geq 0.9995$, the code sets $\theta_1 = \theta_2 = 0$; none of the 24 hold-out refits needed this, and no GARCH margin fit failed. BLAS threads are set to one so that the numbers should not depend on the number of cores; I did not test this on another machine.

10.4 Reproducing the results

Table 10.1 gives the three levels of reproduction. The quickest rebuilds the thesis from the stored result files; the slowest recomputes everything from the raw data.

`build_all.py` runs `thesis_data_summary.py`, `thesis_tables.py`, `thesis_fig_`

Table 10.1: How to reproduce the results

Task	Command (from the repository root)	Time
Rebuild all tables and figures, check every registered number, compile the PDF	`python scripts/build_all.py` (add `--no-pdf` to skip LaTeX)	seconds
Run the tests	`python -m pytest -q`	about 1 minute
Post hoc checks and baselines	`python scripts/thesis_posthoc.py`, `thesis_baselines.py`	seconds
Dry run and hold-out	`python scripts/confirm_00_run.py pseudo` or `holdout`, then `confirm_01_score.py`	under 1 minute
Development results	`scripts/explore_00` to `explore_20` in order	several hours
Power at the hold-out length	`scripts/explore_21_power_holdout_length.py`, then `explore_22_power_summary.py`	about 19 s per replication, 2,000 replications

Note: The hold-out runner refuses to run unless the frozen files match the tag and refuses to overwrite an existing output; the stored hold-out forecasts are `output/confirm/holdout_forecasts.pkl`. `scripts/verify_clean_clone.py` repeats the build in a fresh clone and compares each table, figure and the PDF text with the committed versions.

`forest.py`, `thesis_fig_cumulative.py`, `thesis_fig_regime.py` and `check_numbers.py`, then compiles the LaTeX source with `tectonic`. Every number quoted in the prose that comes from a result file is registered in `thesis/numbers.yaml` with its source file and format, and `check_numbers.py` recomputes each one and checks that it appears in the stated chapter. Each table and figure names its result file and script in its note, and the repository's README collects the same map in one table.

10.5 Checks on the code

The tests check the parts of the pipeline on which the results depend: that no forecast uses information from its own day or later and that dates align across assets; that the FZ0 loss is lower at the true VaR and ES than at biased ones on simulated normal data; that the mixture VaR and ES match their closed forms; that the hidden Markov model recovers known states; the sign convention of Z_2 and the Christoffersen test; the scorer's decision rule and validity checks; and the hold-out lock. The scored hold-out tables are listed with their SHA-256 hashes in `output/confirm/holdout/manifest.json`, and the forecast file has SHA-256 `9537bb5d6008028076126d0166bb2edc1f643e5a9f496cb852d7872f8485f974`.

Chapter 11: Reflection

11.1 From a broad question to a single test

The project began in Phase 1 with a broader question: does a regime-switching covariance model, in which volatilities and correlations both depend on a market state, forecast the tail risk of a stock–bond–gold portfolio better than a single-regime model? The pilot built the static models A to D on SPY, IEF and GLD and judged them against six go/no-go criteria that I wrote down before running them, covering correctness, the size and power of the correlation test, the stability of the estimated regimes, reproducibility and run time. All six passed, but two only narrowly: the power to detect a correlation breakdown reached 50% only in the strongest simulated scenario, and the regimes were stable in 90.3% of refits against a 90% bar. I proceeded with the design, but these margins were the first sign that the correlation question would be hard to answer with daily data on three funds.

My supervisor then asked for a broad exploration of regime definitions and types of ETF. I split the sample so that 2005–2019 would serve for development and 2020–2025 would stay untouched until the test was fixed, and I logged every specification I tried. The exploration changed the question in two ways. First, the static comparison of regime-specific against pooled correlation, which the pilot had judged correctly sized, turned out to be oversized once volatility follows GARCH dynamics, so its many raw rejections were not evidence. This moved the question to models with GARCH margins, where the correlation channel could be isolated with the volatility model held fixed. Second, the long history of fund proxies showed that the development period contains no spell of positive stock–bond correlation, so only the hold-out could speak to the 2022 reversal. The question narrowed from "do regimes help?" to one comparison of one correlation, fixed in advance.

11.2 Challenges and how I addressed them

The first challenge was theoretical framing. The literature disagrees about whether the stock–bond correlation follows a slow macroeconomic state or market stress, and my regime signal, built from equity volatility, can speak only to the second. I chose to state this openly as the reading the test addresses rather than to claim more, and Chapter 8 reads the result against both.

The second was methodological. Comparing nested forecasting models out of sample is not covered cleanly by textbook tests, and the papers on the subject treat linear models under squared-error loss. Reading them led me to define the estimand as the expected loss difference of two forecasting procedures, in the sense of Giacomini and White, and to check the size of the test by simulation rather than rely on asymptotic theory. The power study then showed that six years of hold-out data would detect a moderate correlation shift only about one time in five. Rather than look for a more favourable test, I kept the one declared and added a smallest effect of interest and minimum detectable differences, so that a non-significant result would still say something.

The third concerned how results are worded. My supervisor's feedback on an earlier design document cautioned against overstating what each comparison isolates and against reading a non-significant result as evidence of no effect. I corrected that document, built Table 4.1 to state what each comparison changes, and fixed the wording of every possible outcome before the run.

11.3 What the record shows and what I would change

Two episodes tested the discipline of the design: I displayed two single-day hold-out returns by accident while searching the test files, and the scorer refused forecasts recorded two commits after the frozen tag, which I resolved by amending the check before any number was computed (Appendix A). Both episodes showed me that a pre-registration kept inside one's own repository is only as credible as the record around it, which is why the appendix sets them out in full.

With hindsight I would change three things. I would include the practitioner baselines, historical simulation and an EWMA, in the frozen design, since the simplest of them turned out to be the best calibrated. I would choose the regime signal from the economic argument, for example from inflation or from the stock–bond correlation itself, rather than from the exploration and the power study. And I would treat calibration as a primary outcome alongside the loss comparison, because for a risk desk it was the more consequential result. What I would keep is the order of work: explore freely, write the test down, then use the hold-out once.

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Appendix A: Disclosures and Run Record

A.1 Items seen before the specification was fixed

Besides the two items discussed in Section 3.4.2, the following were seen or used before the specification was fixed.

1. A descriptive data analysis and a data audit in the pilot displayed summary statistics for 2010 to 2025. Calendar-year SPY–IEF correlations, Table 3.1, Figure 3.1 and the reference baselines of Section 7.4 were computed after the hold-out was scored, for description only.

2. Three unit tests read the 2010–2025 pilot returns and passed in earlier sessions; they were excluded from the test run until the hold-out had been scored.

3. A search of the test files printed two single-day returns, for 2020-03-10 and 2023-11-01, once by accident. They were not used, and no specification depends on them.

A.2 The record of the run

The forecasts were produced once by `scripts/confirm_00_run.py` from the code fixed by the tag `spec-freeze-v1`, for 1,508 forecast days from 2 January 2020 to 31 December 2025. Before the run the file hashes in the specification matched the files, the tests passed, and no earlier hold-out output existed. The supervisor had been sent the hypothesis, the sidedness and the exact SPY–IEF specification, and had replied that the test looked appropriate.

The scorer then refused to score. Its first check required the commit recorded in the forecast file to equal the tagged commit, and the recorded commit was two commits later: those commits changed an ignore file and a log, and none of the forecast code. No number from the run had been computed or seen. I replaced the check with one that requires the recorded commit to descend from the tagged commit and the forecast code to be identical in both, wrote the change and its reason into the specification (its section 8), and tagged it `spec-freeze-v1.1`. The forecasts were not produced again. The original rule was part of the specification, and the amendment,

although it cannot depend on a result that did not yet exist, is a departure from it. It also goes beyond the letter of the specification, which allowed a repair only after a crash before output was written, whereas here the output existed and the scorer refused it. The new check covers `src/`, the runner and the requirements file; it does not cover configuration files, the other scripts or the Python environment. That the amendment was made before any figure was seen rests on my record and on the commit times, not on an independent witness.

A.3 Frozen wording of the other outcomes

The wording fixed before the run for the outcomes that did not occur was:

O1 "The hold-out data provide statistical evidence at the one-sided 5% level that letting the SPY–IEF correlation switch with the regime lowers expected FZ0 loss relative to constant correlation (difference $\bar{d}$, 95% CI $[a, b]$). This is one test on one six-year sample containing a single stock–bond reversal; simulated power was P%."

O3 "The data are inconsistent, at the one-sided 5% level, with an improvement larger than SESOI $= 0.002$ FZ0 units. Improvements smaller than this, and any degradation, are not excluded."

O4 "The hold-out data provide evidence that the switching specification has higher expected FZ0 loss ($\bar{d}$, 95% CI $[a, b]$)."

Appendix B: Mean-Treatment Controls

The development comparisons of Chapter 6 use the pooled window mean in every component. Figures B.1 and B.2 repeat Figure 6.1 with a zero mean and with regime-specific means. The signs of the volatility and margin contrasts do not change, but their sizes do: the RSC-versus-D contrast is larger in absolute value with a zero mean. The correlation contrasts stay small under every treatment, with the exceptions noted in Section 6.2.3. The full tables, including the sub-periods, are in `output/explore/ladder/ladder_dm_table.csv`.

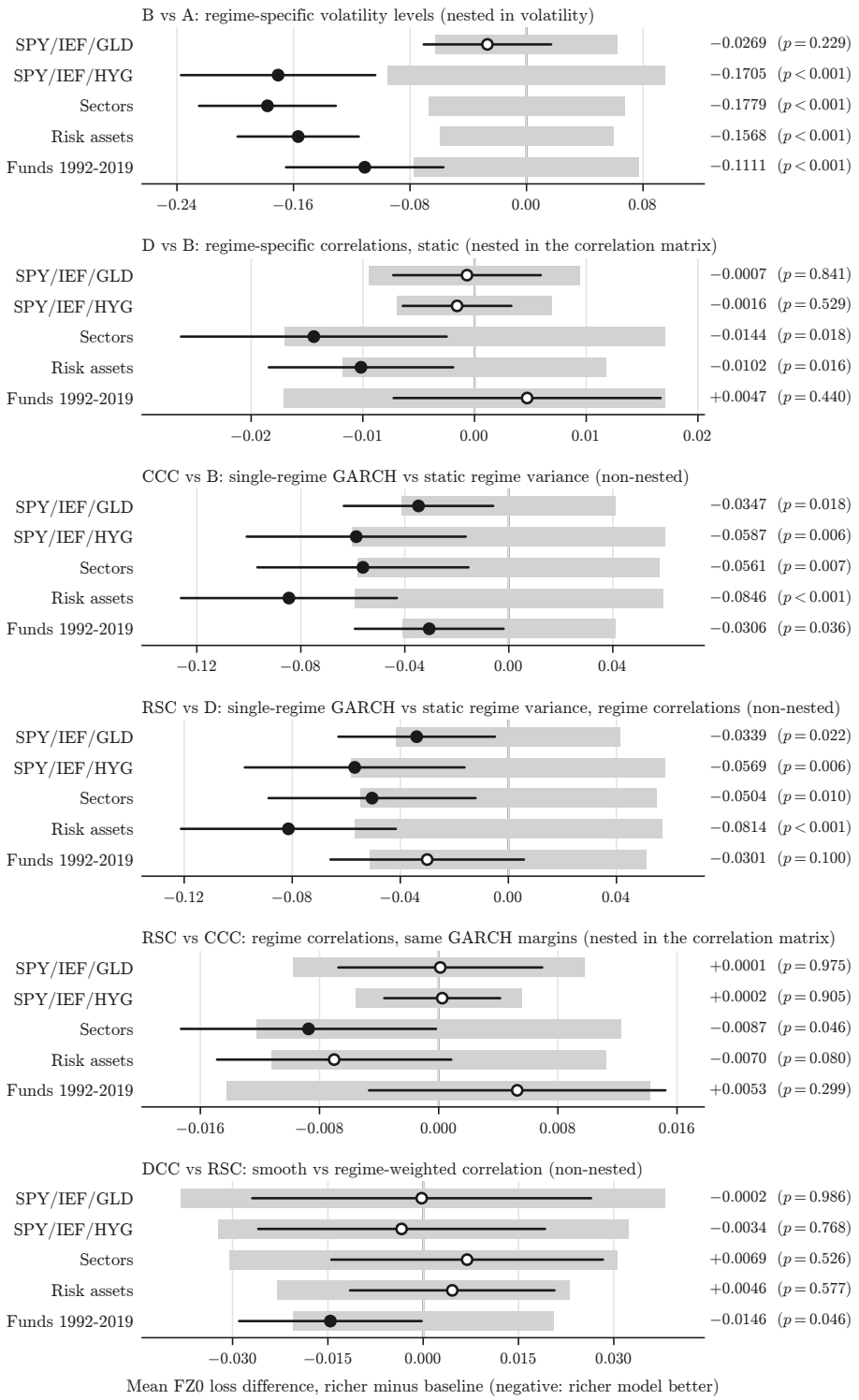


Figure B.1: Mean FZ0 difference by comparison and sample, zero mean in both models

Note: As Figure 6.1, with the forecast mean set to zero in every component. Source: `output/explore/ladder/ladder_dm_table.csv`.

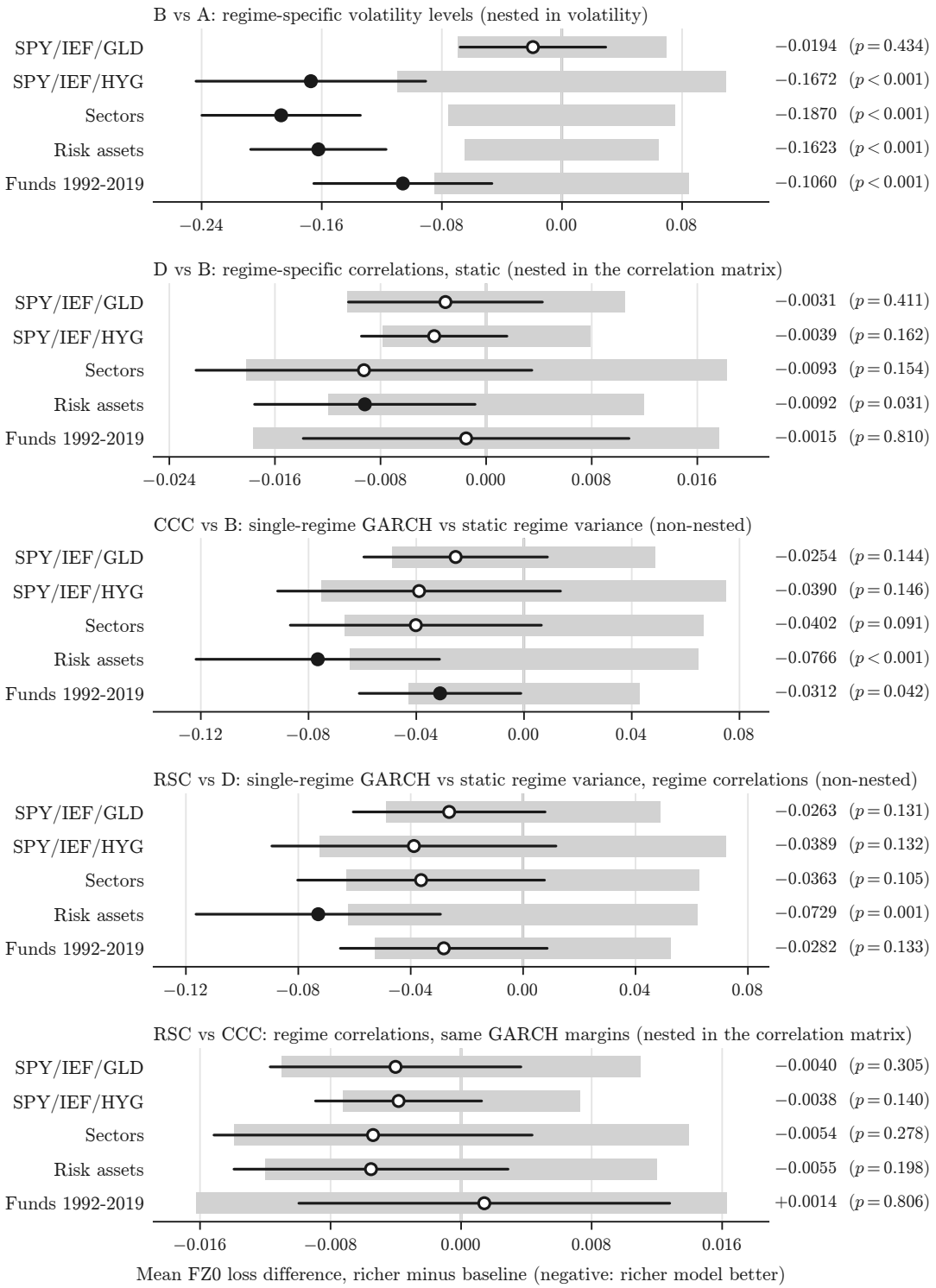


Figure B.2: Mean FZ0 difference by comparison and sample, regime-specific means

Note: As Figure 6.1, with a regime-specific mean in each component of the static models; the GARCH-margin models and DCC have no regime-mean version, so the DCC panel is omitted. Source: `output/explore/ladder/ladder_dm_table.csv`.

Appendix C: Technical Details

C.1 Weighted moments of the static models

For weights $q_s \geq 0$ over the window days, the weighted mean and covariance of the asset returns are

$$\hat{\mu}(q) = \frac{\sum_s q_s r_s}{\sum_s q_s}, \qquad \hat{\Sigma}(q) = \frac{\sum_s q_s \, (r_s - \hat{\mu}(q))(r_s - \hat{\mu}(q))'}{\sum_s q_s}, \tag{16}$$

with the maximum-likelihood normalisation. The pooled moments use $q_s = 1$; the regime moments use $q_s = \gamma_{s,k}$, with a floor of 10^{-12}.

C.2 Expected shortfall of a normal mixture

For a mixture with weights p_k, means m_k and standard deviations s_k, and value-at-risk q from Section 4.3.3, the expected shortfall is

$$\mathrm{ES} = \frac{1}{\alpha} \sum_k p_k \Big[m_k \, \Phi(z_k) - s_k \, \phi(z_k) \Big], \qquad z_k = \frac{q - m_k}{s_k}, \tag{17}$$

where Φ and ϕ are the standard normal distribution and density functions.

C.3 The DCC recursion

In the DCC benchmark the quasi-correlation matrix follows

$$Q_t = (1 - \theta_1 - \theta_2)\bar{Q} + \theta_1 z_{t-1} z_{t-1}' + \theta_2 Q_{t-1}, \tag{18}$$

where $\bar{Q}$ is the sample covariance of the standardised residuals z, and Q_t is rescaled to a correlation matrix. (θ_1, θ_2) are estimated on the window by Gaussian quasi-maximum likelihood every 63 days.

C.4 Degrees of freedom in the Student-t chain

In the secondary Student-t chain, ν is estimated by maximum likelihood on the window's portfolio returns, standardised by an in-window model standard deviation. In the GARCH-margin models this is the constant-correlation standard deviation, so CCC, RSC and RSC_P share one ν; it is not comparable across the static and GARCH families.

C.5 The HAC variance estimator

The standard error of $\bar{d}$ in Section 5.2.2 is $\widehat{\mathrm{SE}} = \sqrt{\widehat{V}/T}$, with the Bartlett-kernel estimator

$$\widehat{V} = \hat{\gamma}_0 + 2\sum_{\ell=1}^{m}\Big(1 - \frac{\ell}{m+1}\Big)\hat{\gamma}_\ell, \qquad \hat{\gamma}_\ell = \frac{1}{T}\sum_{t=\ell+1}^{T}(d_t - \bar{d})(d_{t-\ell} - \bar{d}). \tag{19}$$

C.6 Parameters of the simulated process

Table C.1: Parameters of the process used in the power study

	Equity-like	Bond-like	Gold-like
Daily volatility, calm state	0.75%	0.30%	0.90%
Daily volatility, stress state	2.0%	0.5%	1.6%
	Equity–bond	Equity–gold	Bond–gold
Correlation, calm state (all scenarios)	−0.30	0.05	0.30
Stress state, S0 (no change)	−0.30	0.05	0.30
Stress state, S1 (moderate breakdown)	0.00	0.20	0.30
Stress state, S2 (strong breakdown)	0.40	0.40	0.30
Stress state, S3 (flight to quality)	−0.55	−0.10	0.40

Note: Persistence of the calm state 0.99 and of the stress state 0.97. Innovations are multivariate Student-t with six degrees of freedom. Each asset's volatility also follows GARCH(1,1) dynamics with coefficients 0.08 and 0.90 on the regime-scaled return, fixed and not estimated. Sources: `config/pilot.yaml` (volatilities, persistence, correlations, degrees of freedom) and `src/dgp_garch.py` (GARCH coefficients).